Data Communications

Lecture #2

What we have discussed

- Introduction to the course
- Introduction to the Internet

Today…

- What does the Internet look like?
- What happens on the Internet

Internet from a service view

- Internet is an infrastructure
- Provides services to applications
  - Web, VoIP, email, games, e-commerce, social nets, …
- Provides providing interface to apps
  - Hooks allowing app programs to connect to the Internet
  - Provides service options

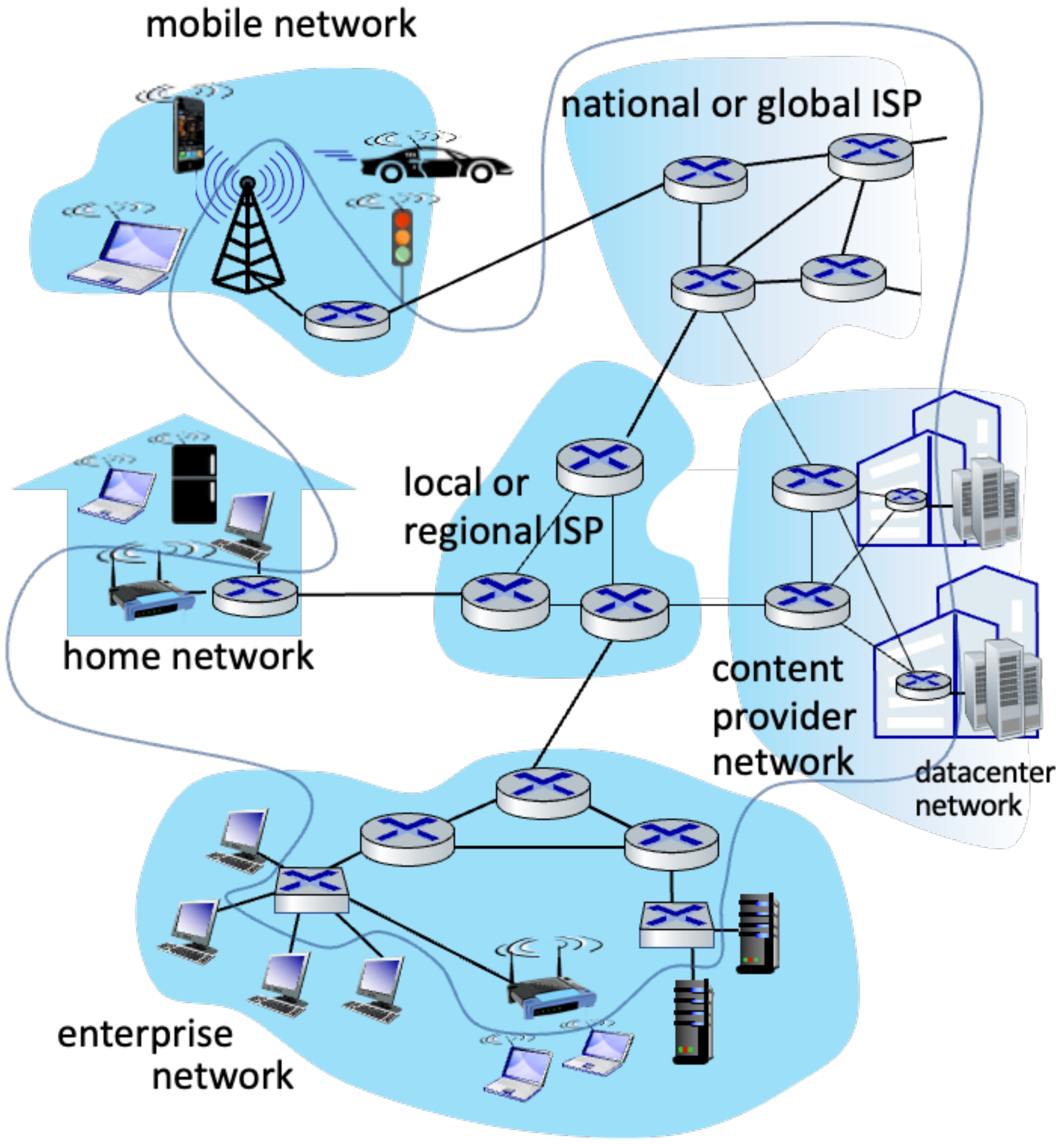
Internet: a “nuts and bolts” view

- Millions of connected computing devices

  - Hosts = end systems
  - Running network apps

- Communication links

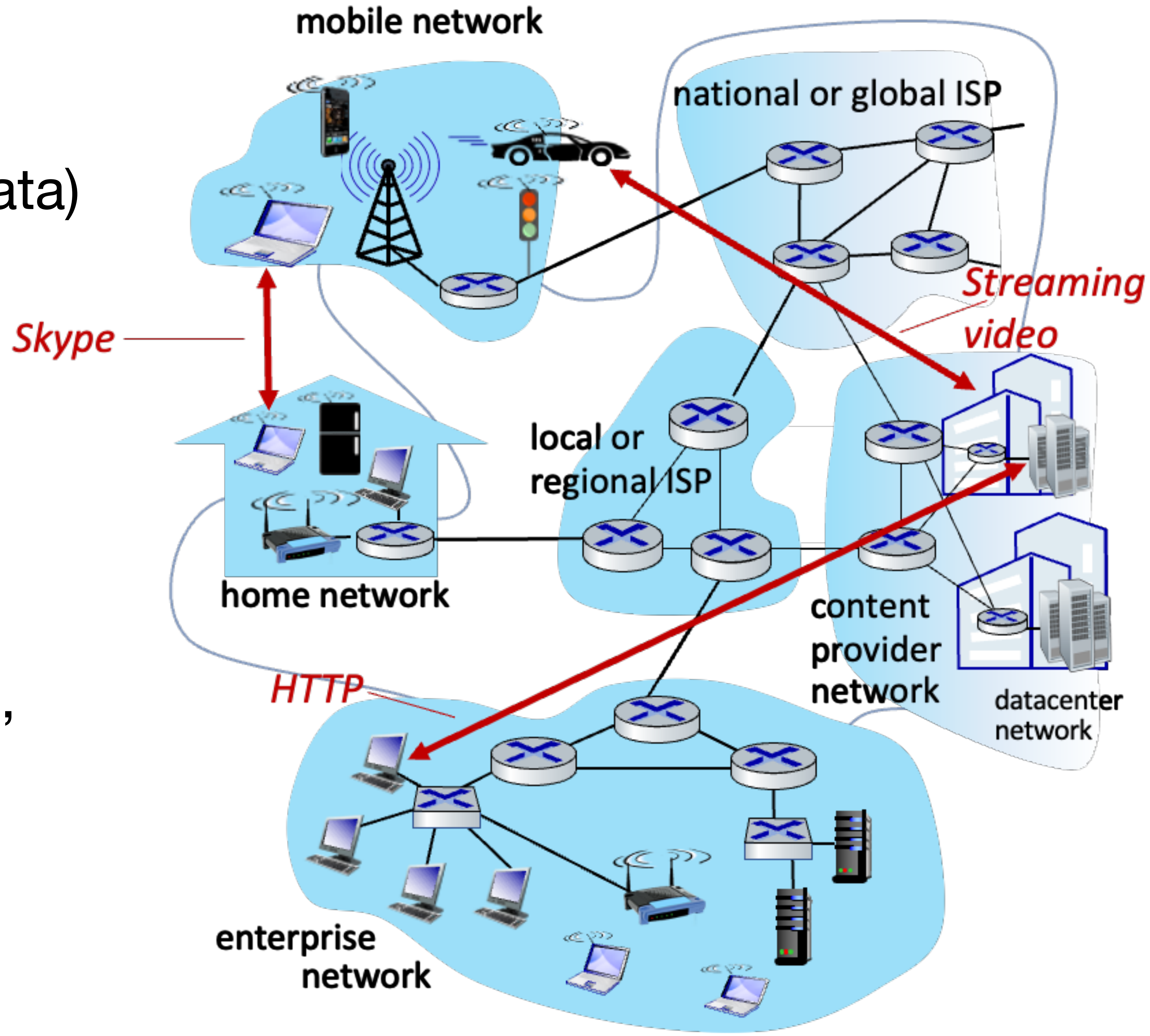
  - Fiber, copper, radio, satellite
  - Transmission rate: bandwidth



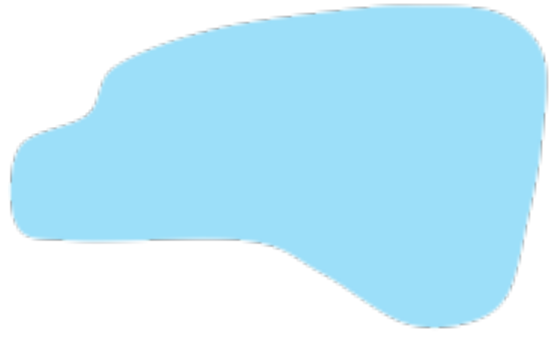
Internet: a "nuts and bolts" view

- Packet switches

  - Forward packets (chunks of data)
  - Routers and switches



- Networks



  - Collections of devices, routers, links
  - Managed by an organization

“Fun” Internet-connected devices

Amazon Echo

Internet refrigerator

IP picture frame

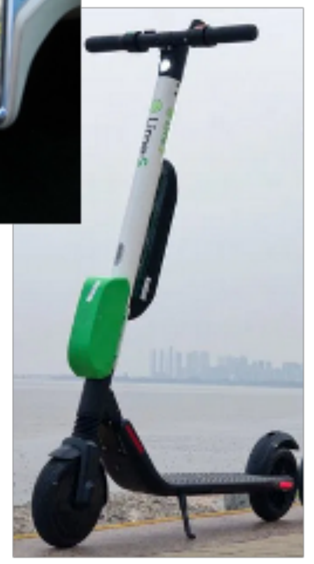
Pacemaker & Monitor

Tweet-a-watt: monitor energy use

bikes

Web-enabled toaster + weather forecaster

cars

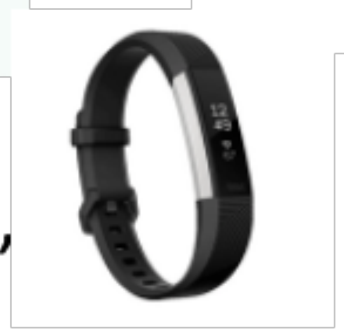


scooters

Security Camera

Slingbox: remote control cable TV

AR devices



Fitbit

Internet phones

Gaming devices

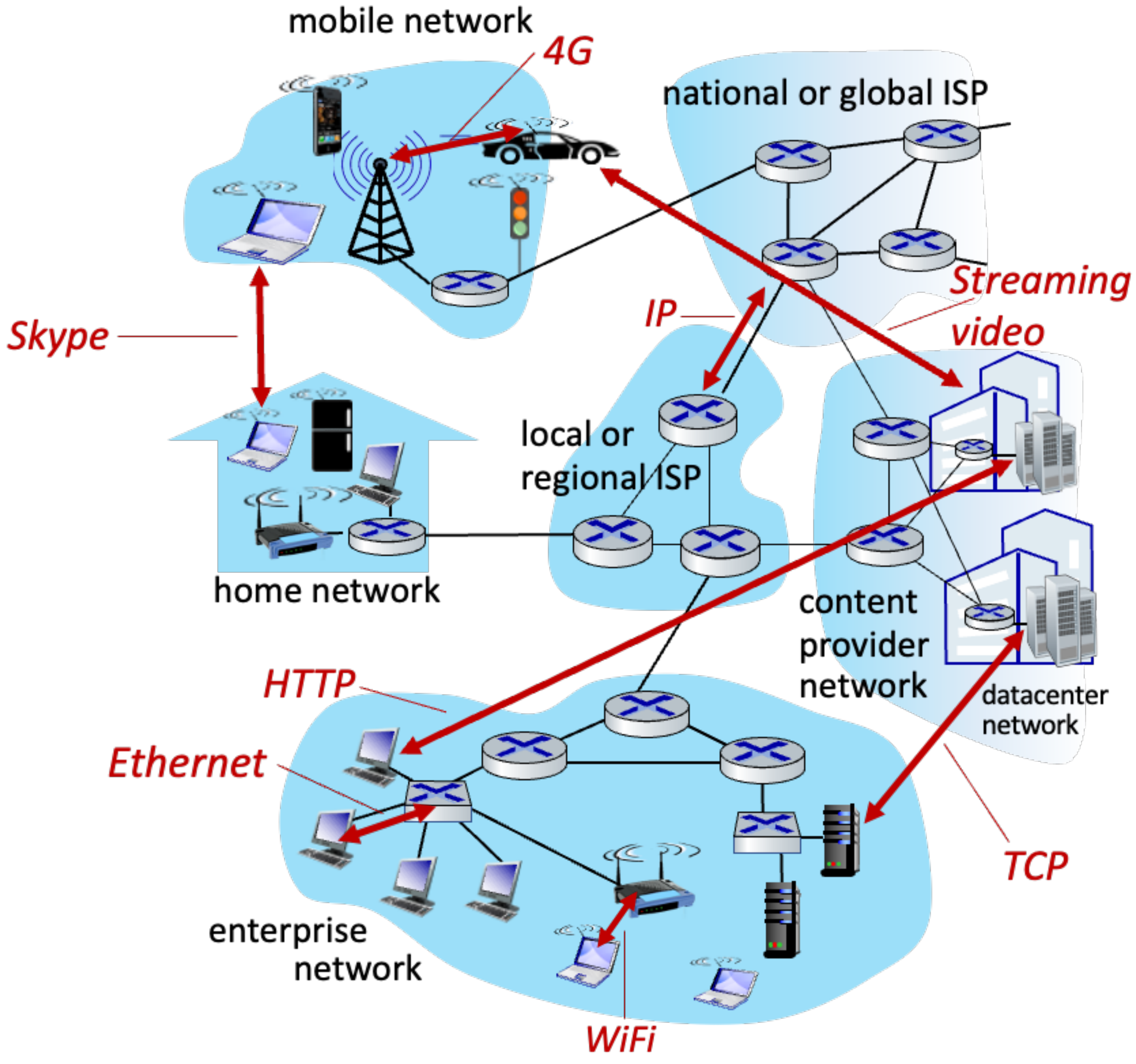
sensorized, bed mattress

diapers

Others?

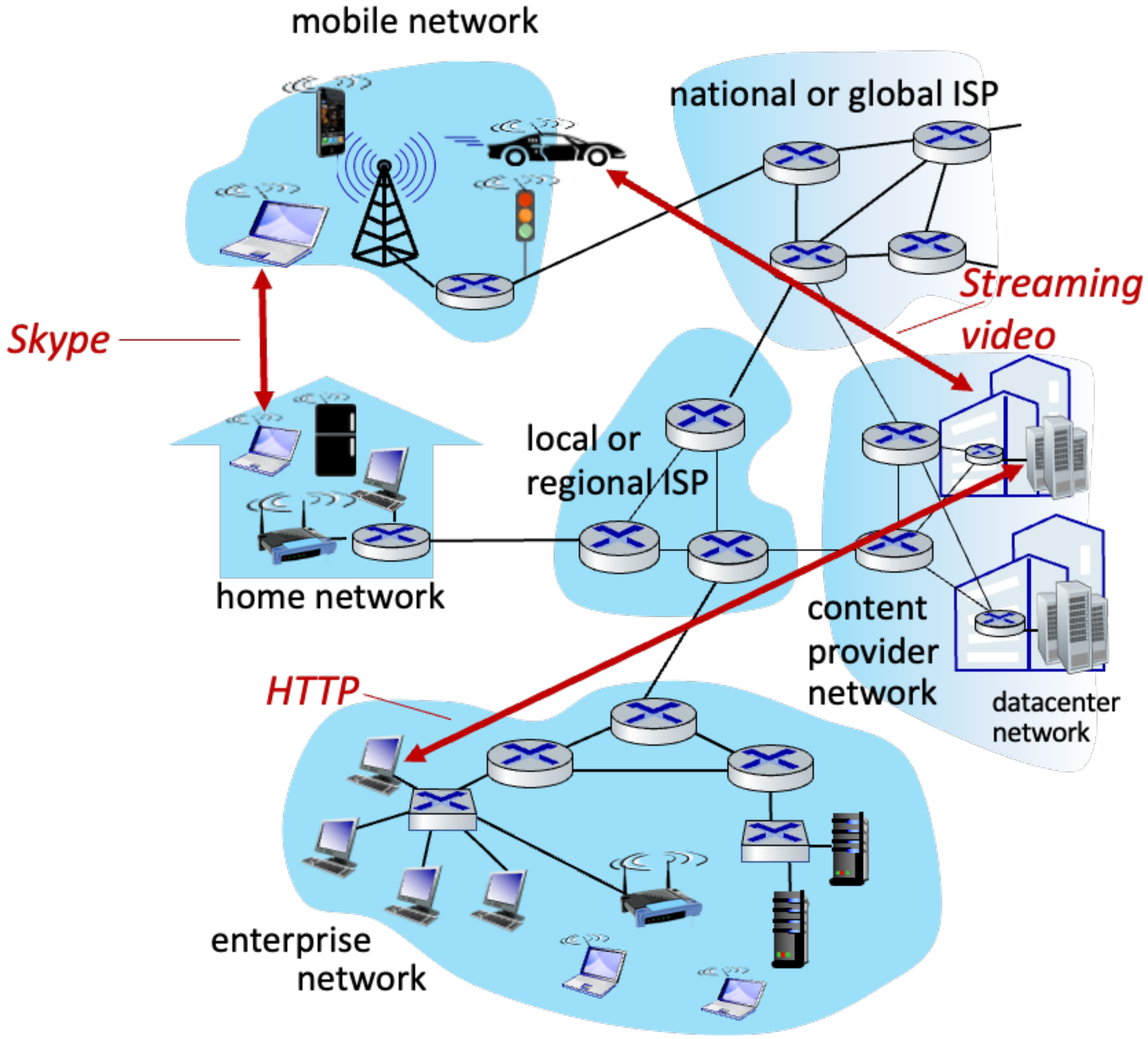
Internet: a "nuts and bolts" view

- Internet: "network of networks"
    - Interconnected ISPs
- Protocols are everywhere
    - Control sending, receiving of messages
    - e.g., HTTP (Web), streaming video, Skype, TCP, IP, Wifi, 4/5G, Ethernet
- Internet standards
    - RFC: Request for Comments
    - IETF: Internet Engineering Task Force



Internet: a "services" view

- Infrastructure that provides services to applications
  - Web, streaming video, teleconferencing, email, games, e-commerce, social media, inter-connected appliances, …
- Provides programming interfaces to distributed applications
  - "Hooks" allowing sending/receiving apps to "connect" to, use Internet transport service
  - Provides service options



What’s a protocol?

- Human protocols
  - What’s the time?
  - “I have a question”
  - Introductions

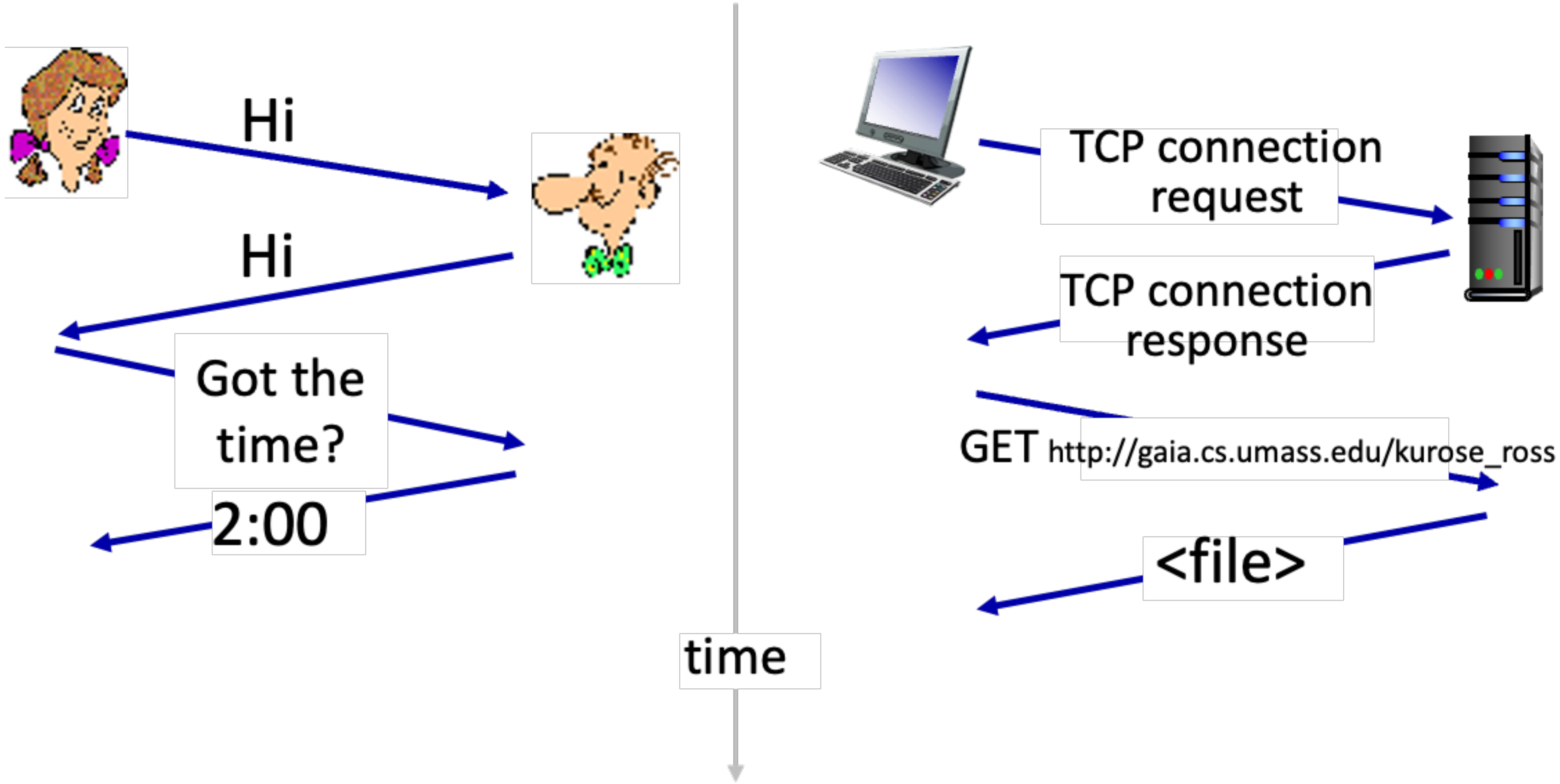
Rules for
… specific message sent
… specific action taken
when message received
or other events

- Network protocols
  - Computers (devices) rather than humans
  - All communication activity in Internet governed by protocols

Protocols define the format, order of messages sent and received among network entities, and action taken on message transmission, receipt

What’s a protocol?

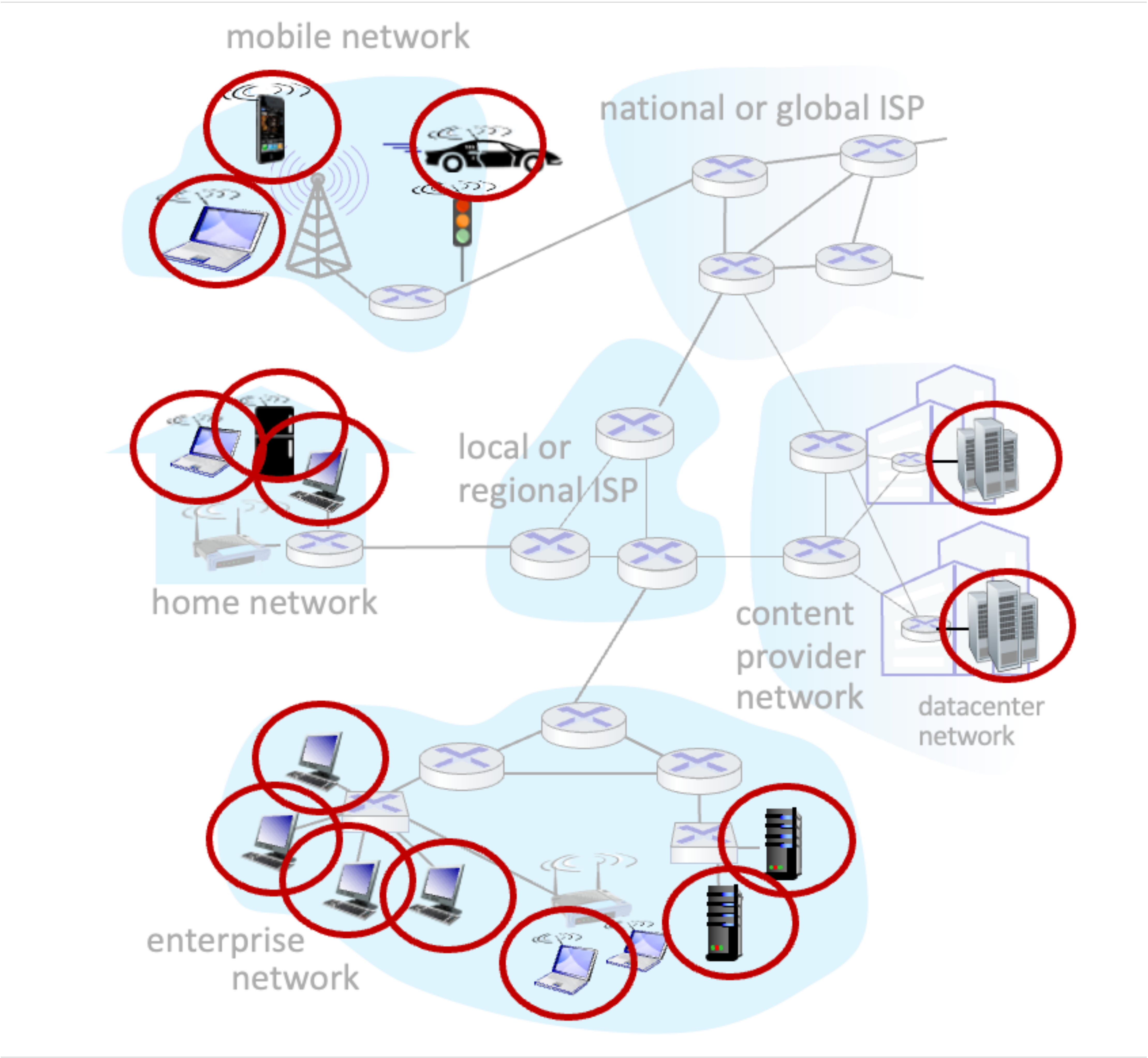
- A human protocol and a computer network protocol



Network Edge

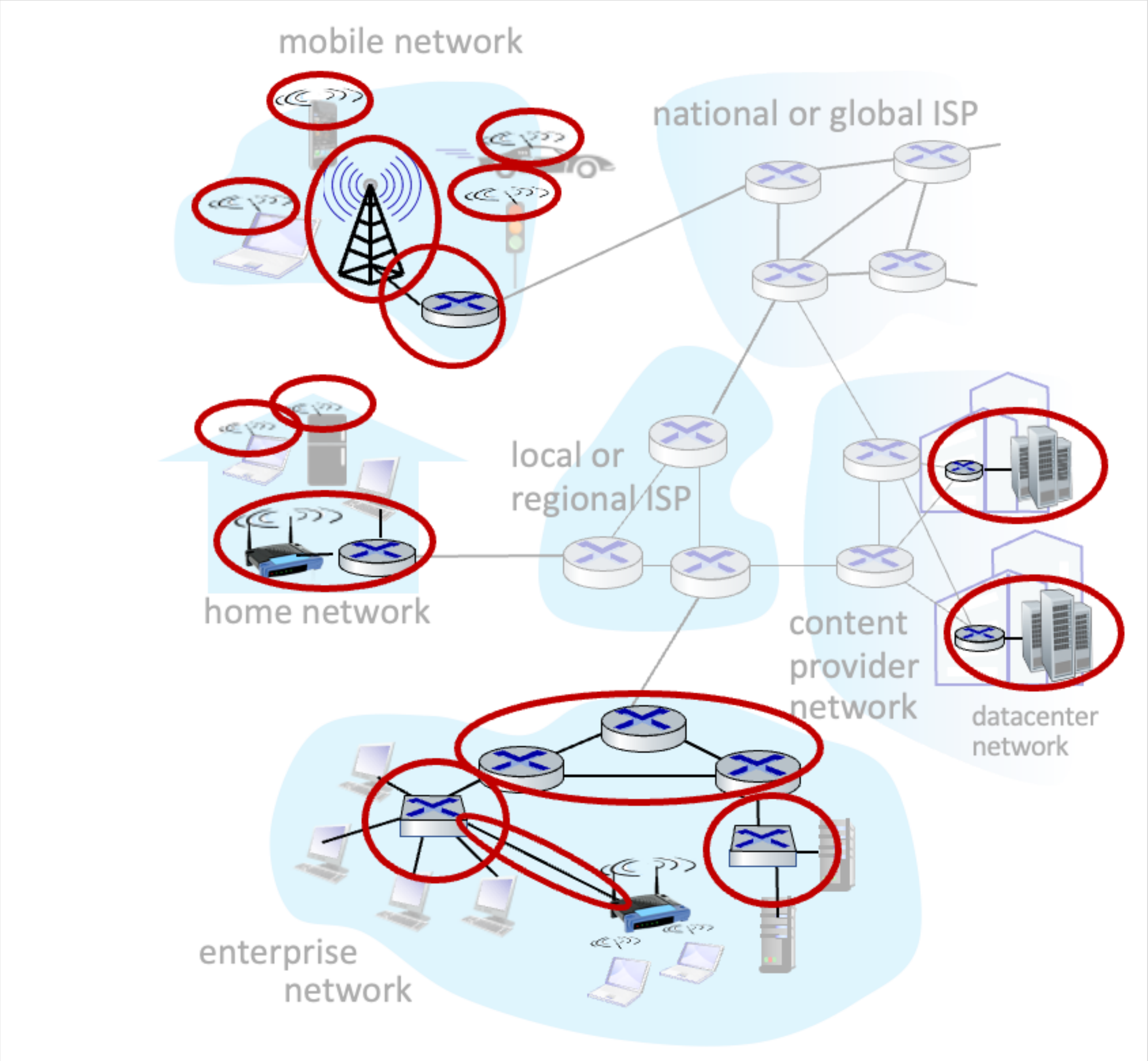
A closer look at Internet structure

- Network edge
  - Hosts: clients and servers
  - Servers often in data centers



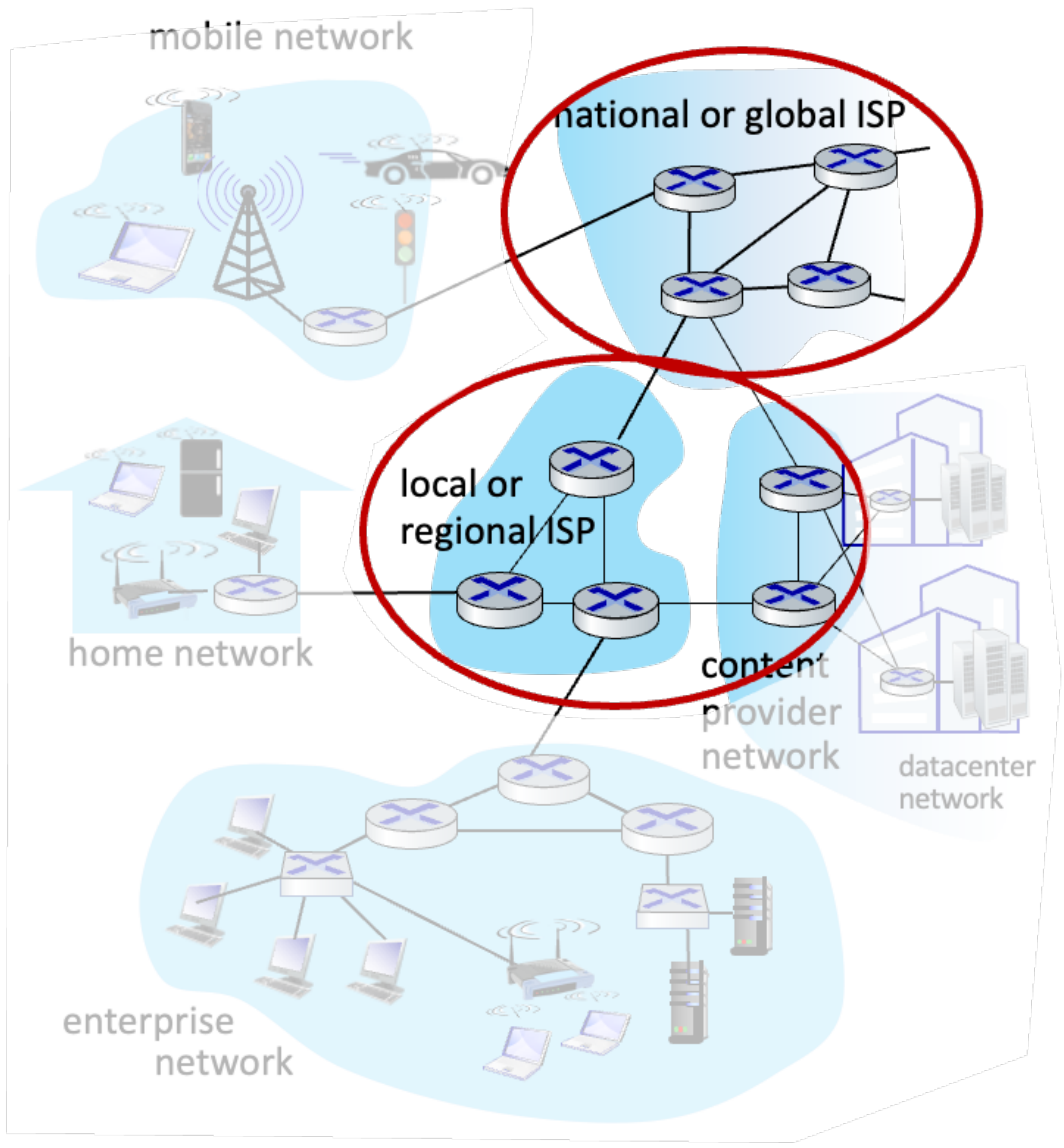
A closer look at Internet structure

- Network edge
    - Hosts: clients and servers
    - Servers often in data centers
- Access networks, physical media
    - Wired, wireless communication links



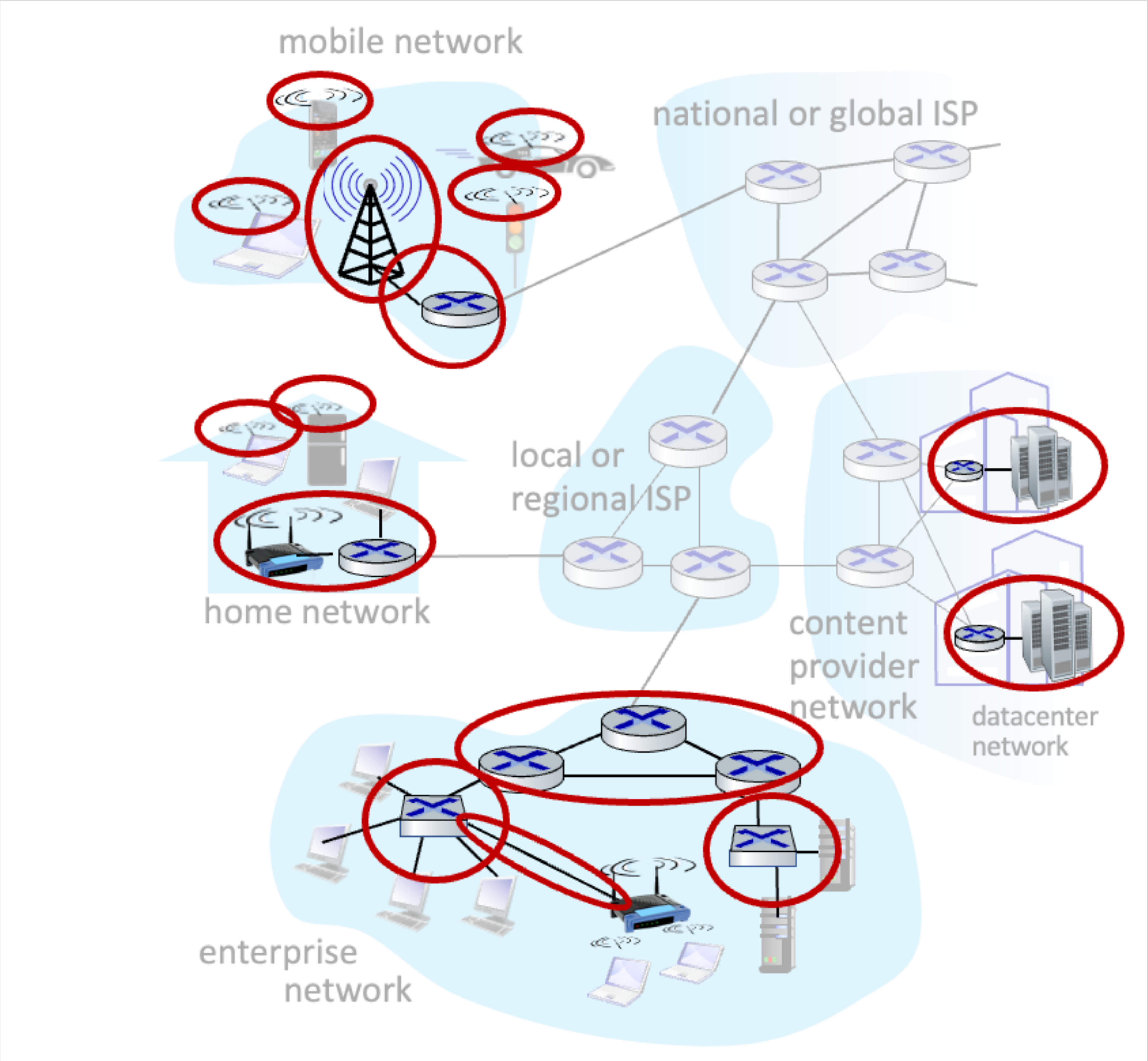
A closer look at Internet structure

- Network edge
  - Hosts: clients and servers
  - Servers often in data centers
- Access networks, physical media
  - Wired, wireless communication links
- Network core
  - Interconnected routers
  - Network of networks

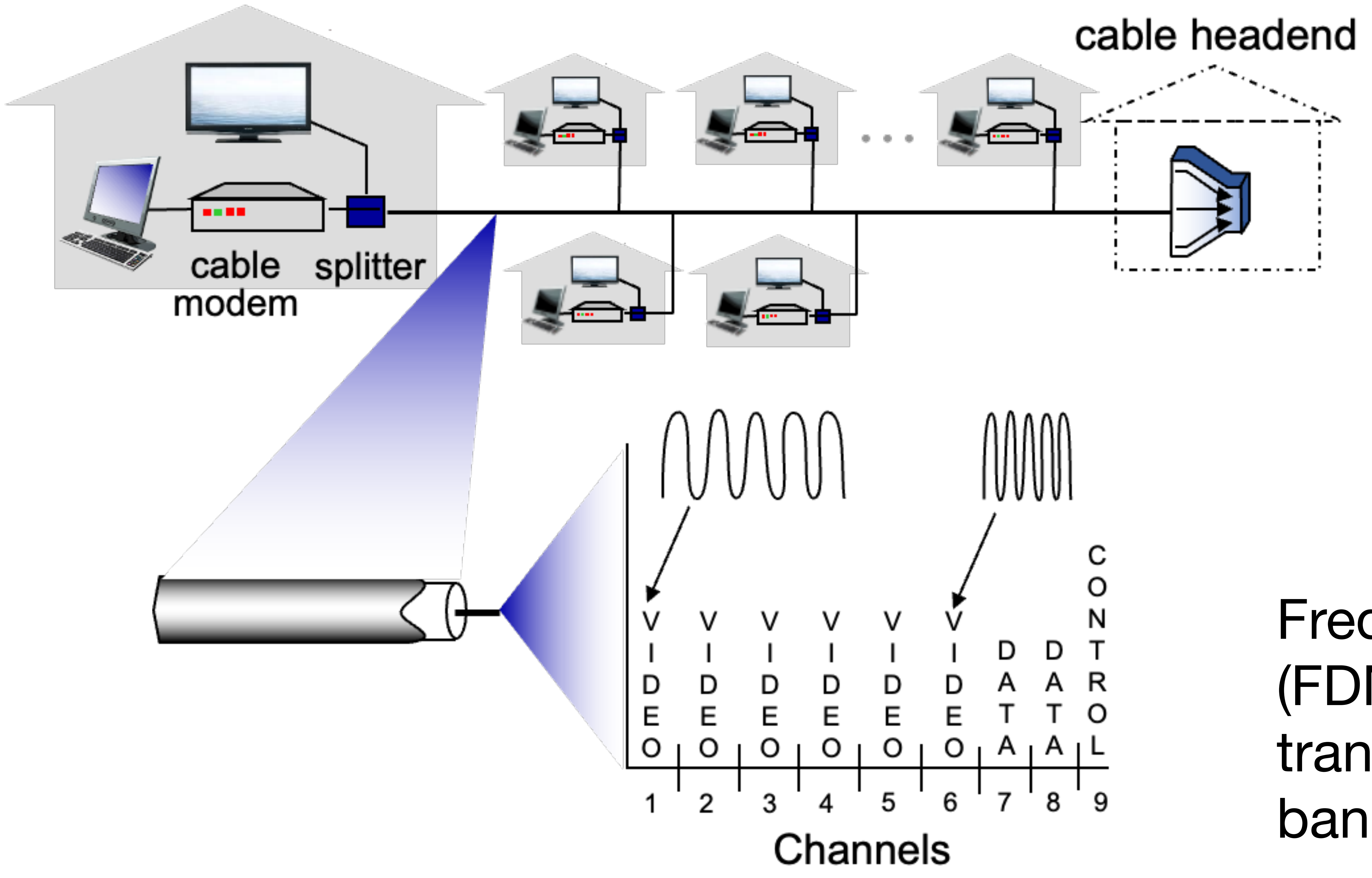


Access networks and physical media

- How to connect end systems to edge router?
  - Residential access nets
  - Institutional access network (school, company, etc.)
  - Mobile access network (Wifi, 4/5G)

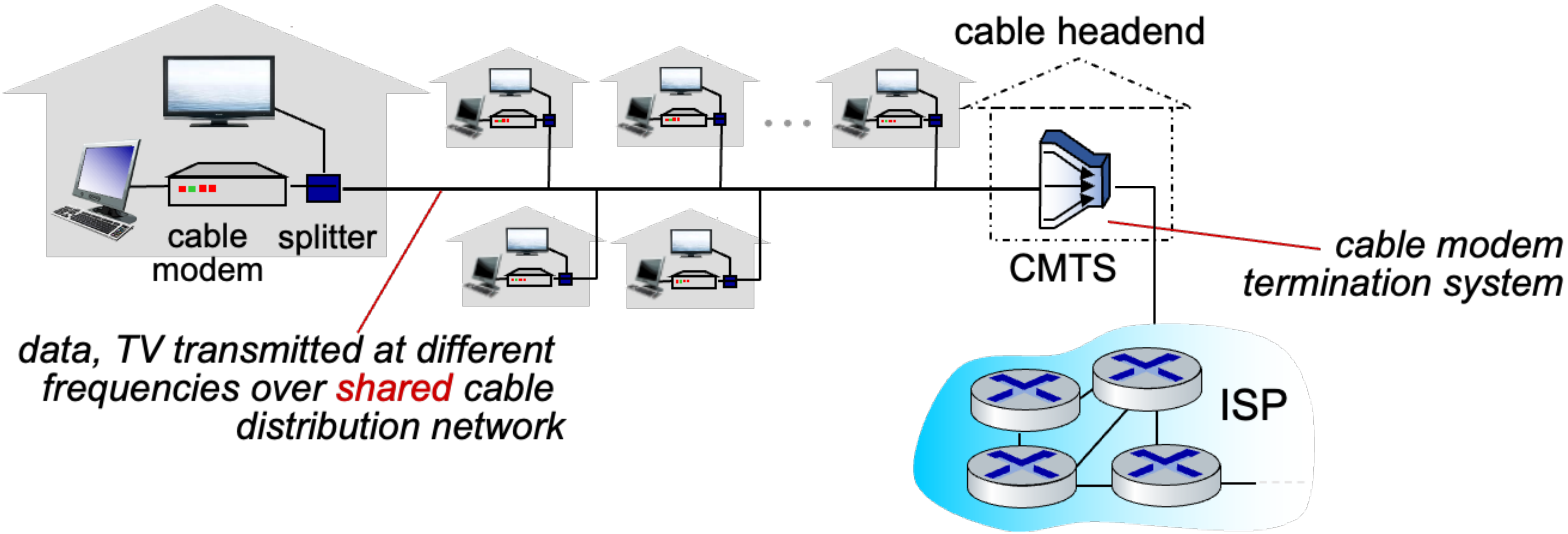


Access networks: cable-based access



Frequency Division Multiplexing (FDM) : different channels transmitted in different frequency band

Access networks: cable-based access

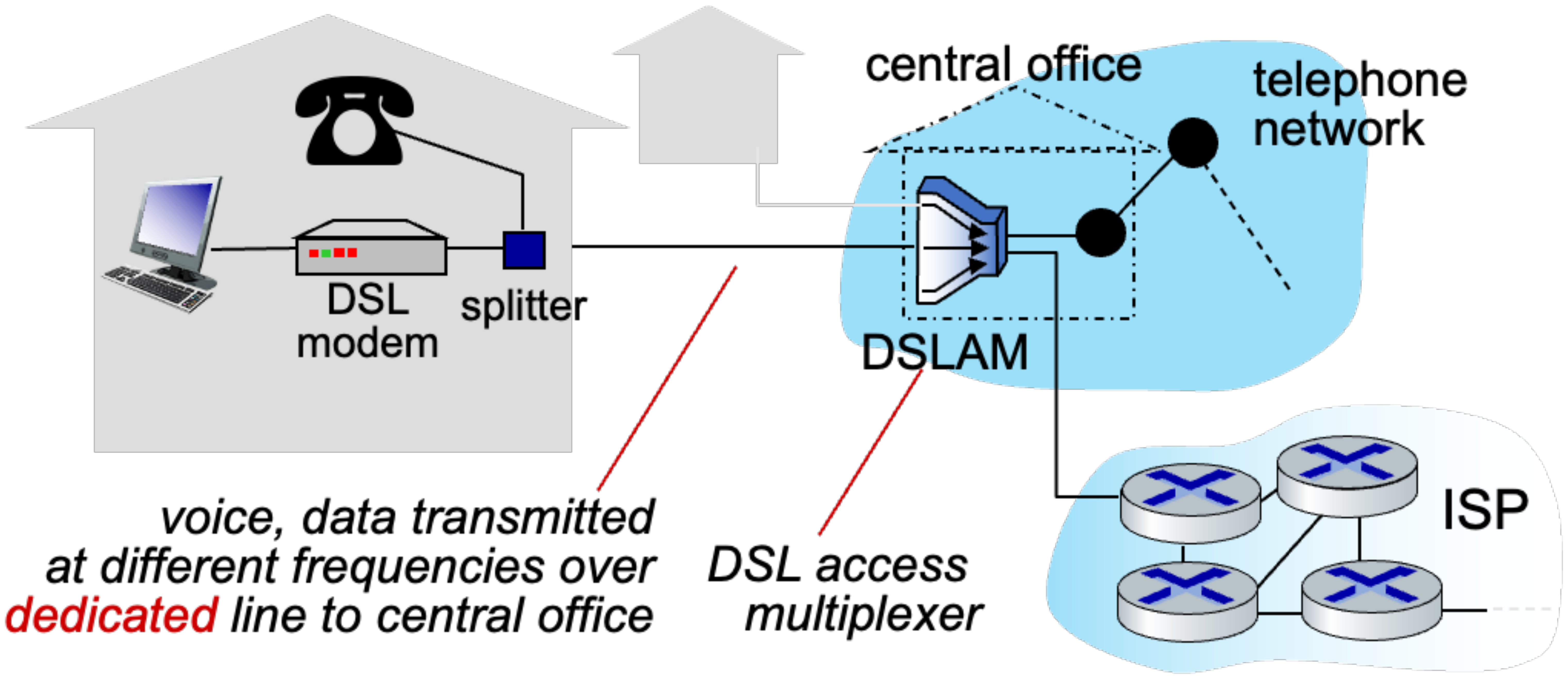


- Hybrid Fiber Coax (HFC): up to 40Mbps~1.2Gbps downstream transmission rate, 30~100 Mbps upstream transmission rate. (Asymmetric!)

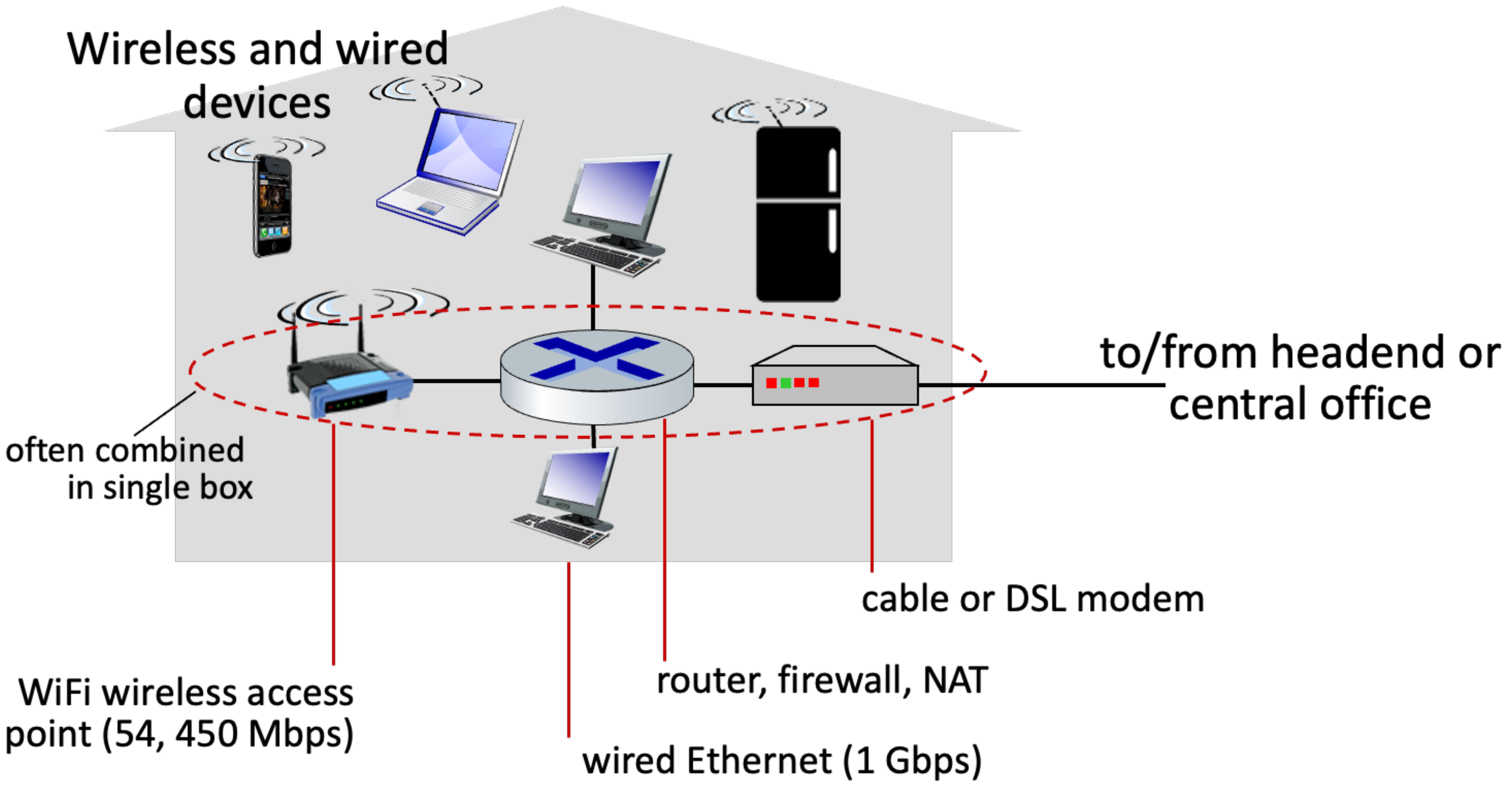
- Network of cable, fiber attaches home to ISP router - Homes share access network to cable headend

Access networks: Digital Subscriber Line

- Use existing telephone line to central office DSLAM
  - Data over DSL phone line goes to Internet
  - Voice over DSL phone line goes to telephone net
- Downstream rate: 24~52 Mbps
- Upstream rate: 3.5~16 Mbps



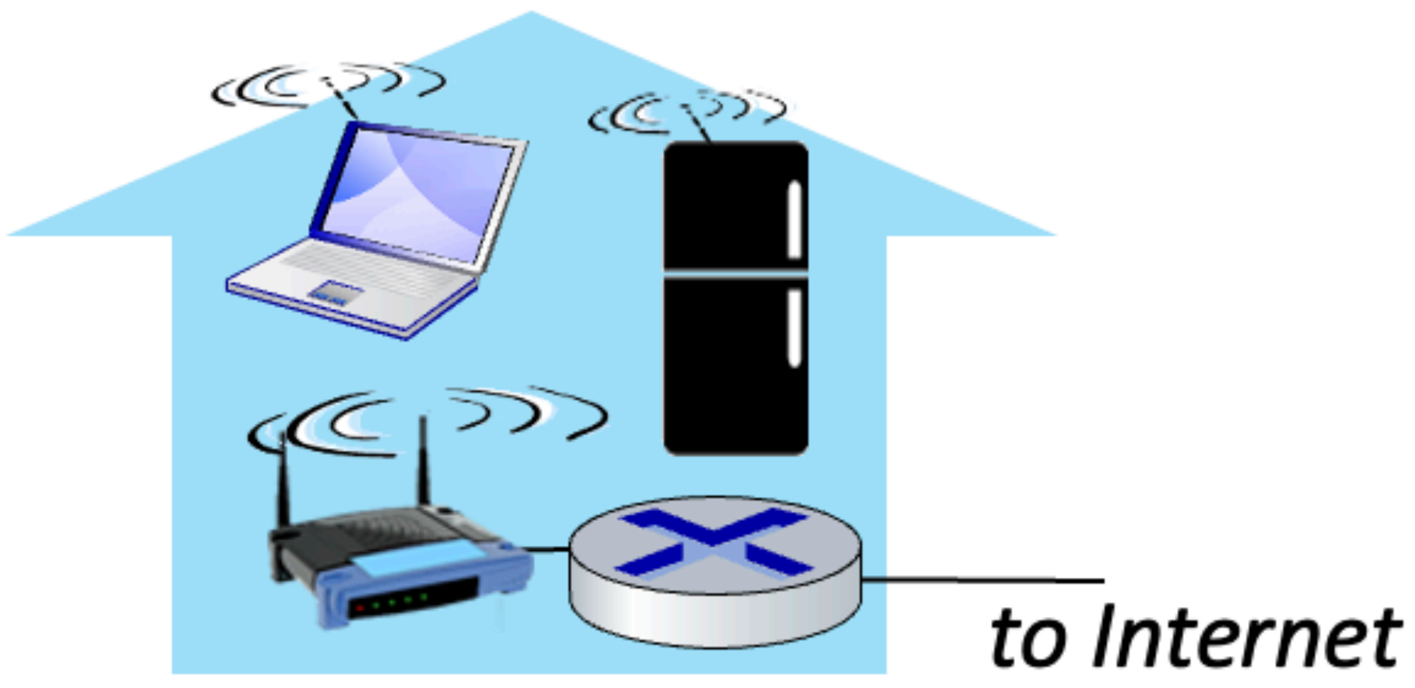
Access networks: Home networks



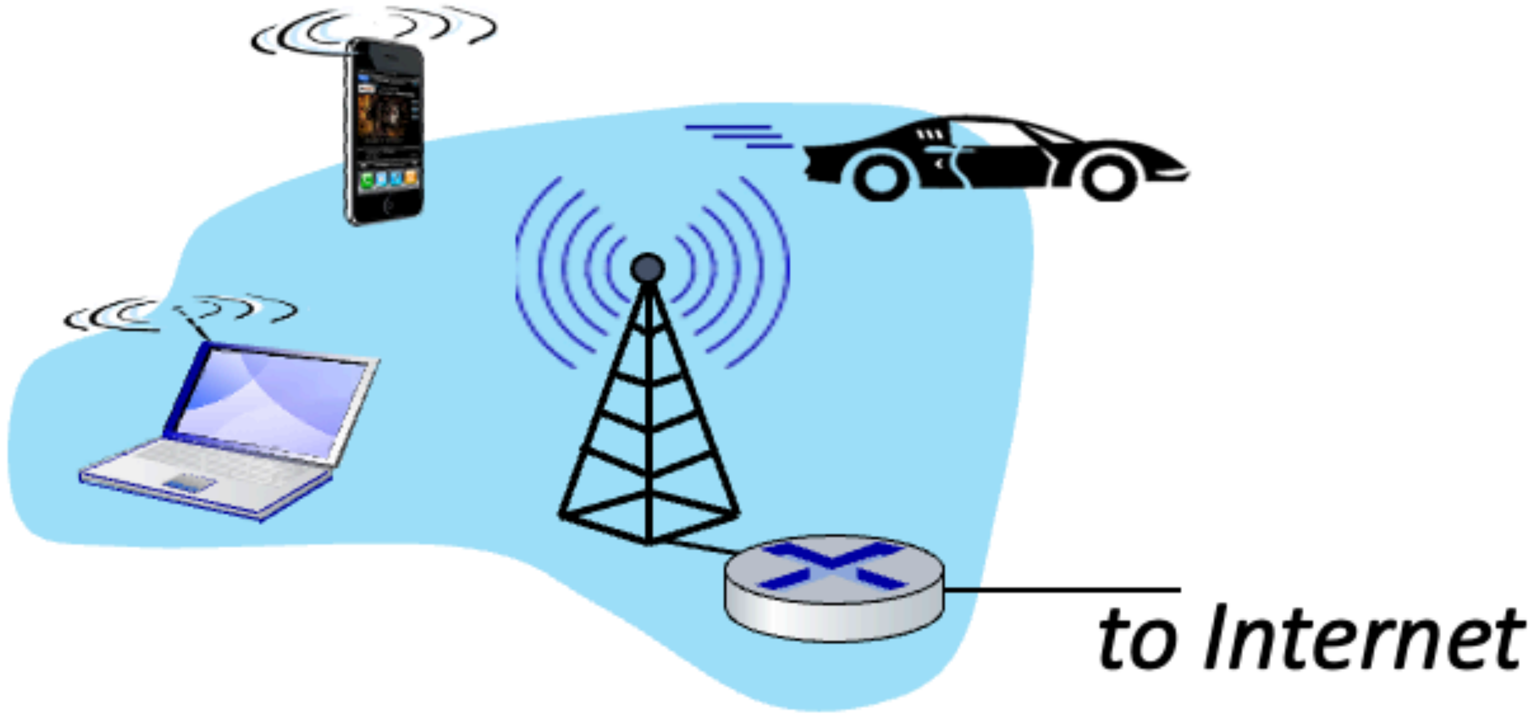
Wireless access networks

- Shared wireless access network connects end system to router
  - via base station aka "access point"

- Wireless local access networks
  - Typically within or around building (~100ft)
  - IEEE 802.11b/g/n (WiFi): 11, 54, 450Mbps transmission rate

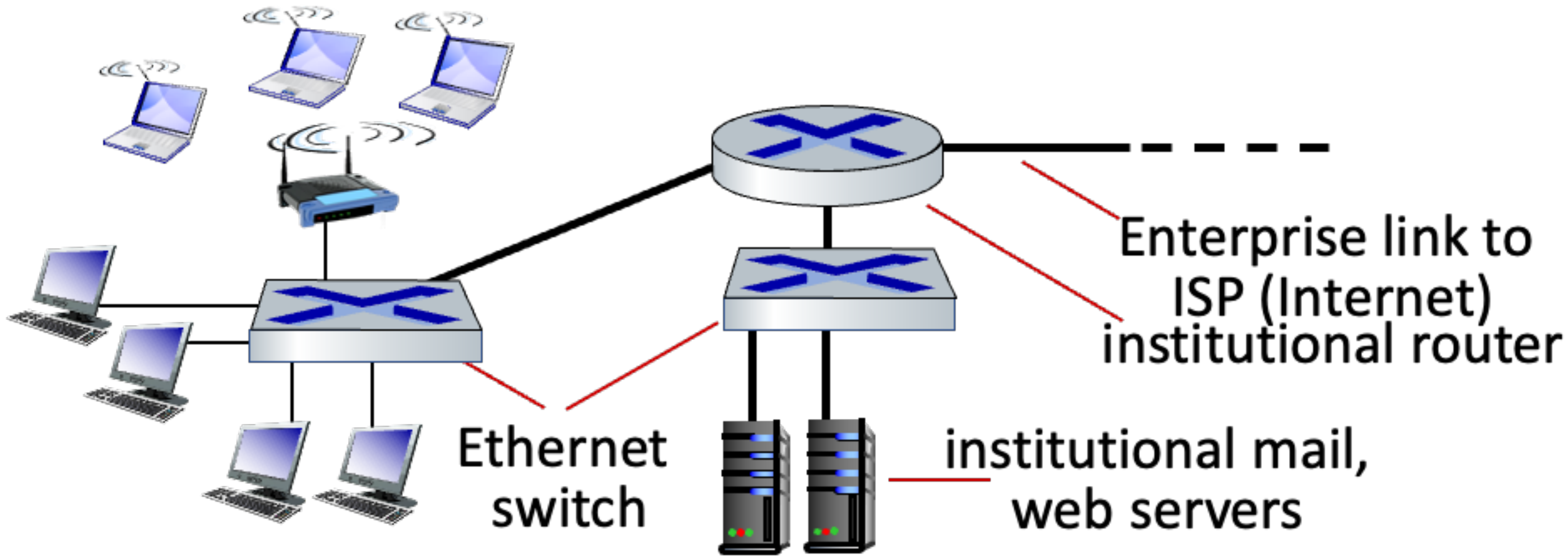


- Wide-area cellular access networks
  - Provided by mobile, cellular network operator (10's km)
  - 4G/5G cellular networks



Access networks: Enterprise networks

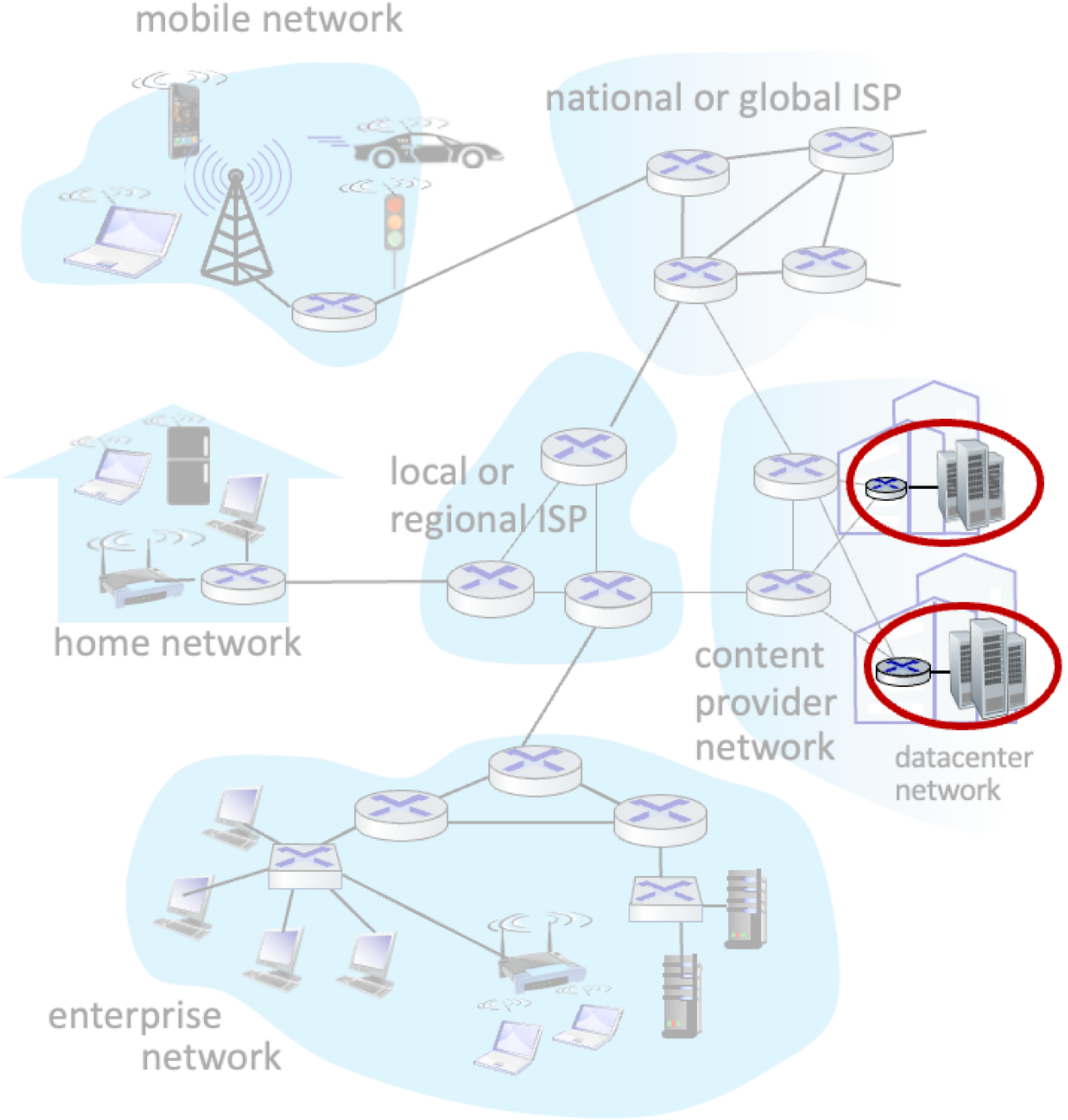
- Companies, universities, etc.
- Mixed of wired, wireless technologies, connecting a mix of switches and routers
  - Ethernet: wired access at 100Mbps, 1Gbps, 10Gbps
  - WiFi: wireless access at 11, 54, 450Mbps



Access networks: Data center networks

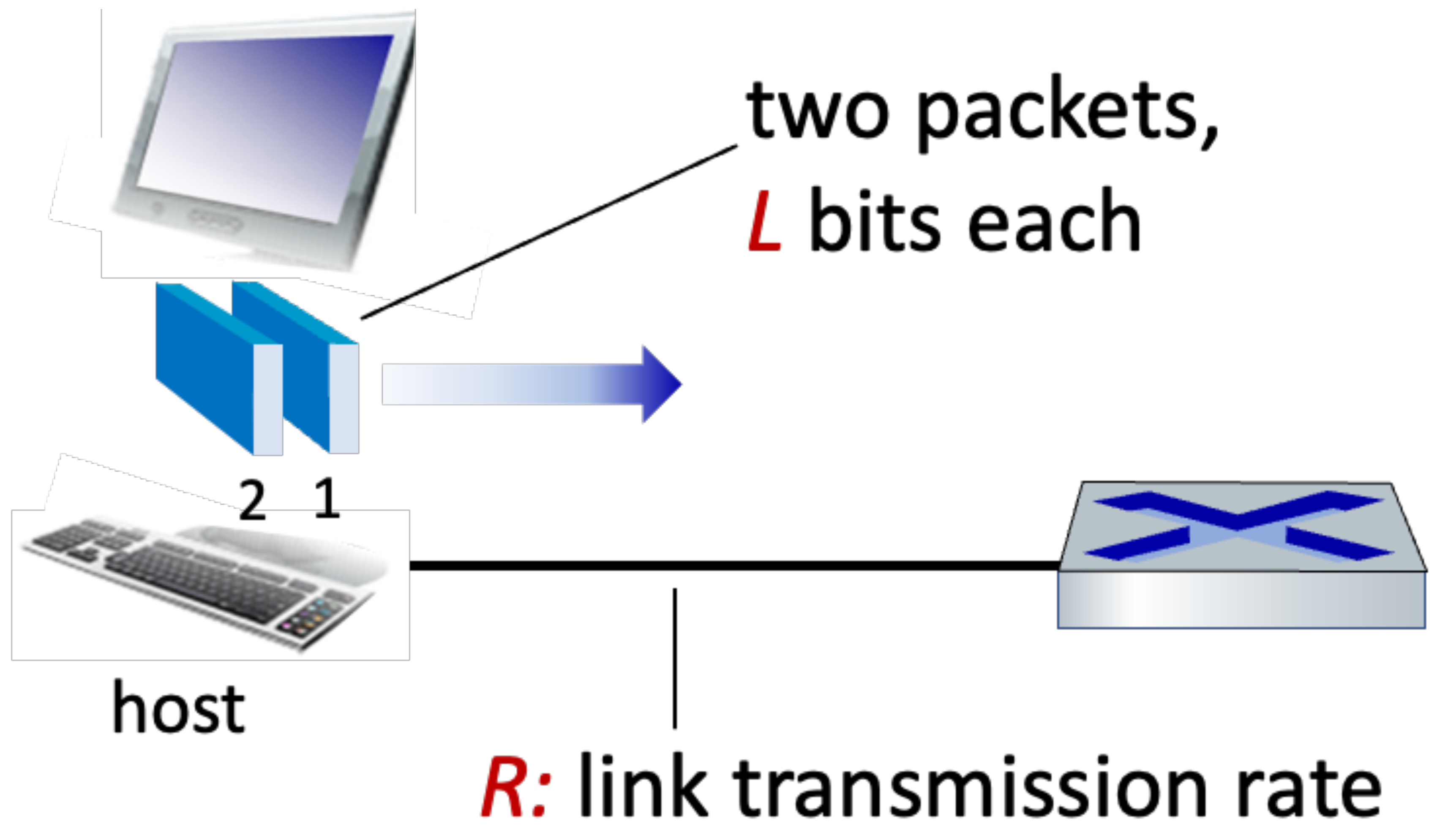
- High bandwidth links (10s to 100s Gbps) connect hundreds to thousands of servers together, and to Internet

Courtesy: Massachusetts Green High Performance Computing Center (mghpcc.org)



Hosts: send packets of data

- Host sending function
  - Takes application message
  - Breaks into smaller chunks, known as packets, of length L bits
  - Transmit a packet into access network at transmission rate R
    - Link transmission rate, aka link capacity, aka link bandwidth



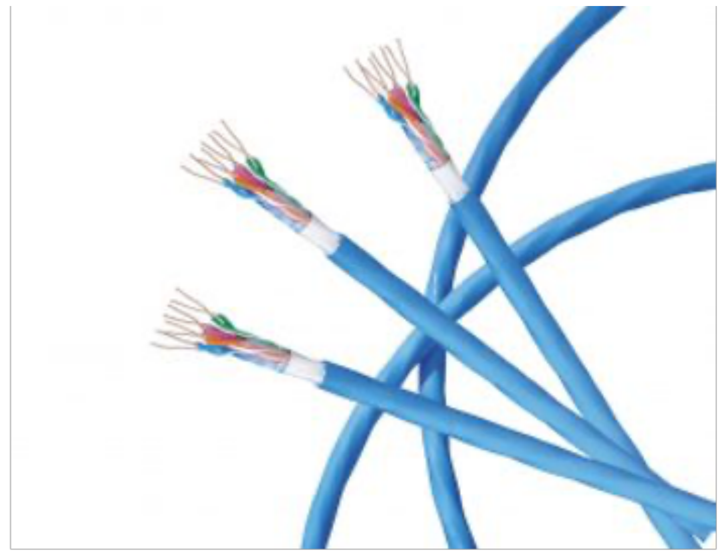
$$\text{packet transmission delay} = \text{time needed to transmit } L\text{-bit packet into link} = \frac{L \text{ (bits)}}{R \text{ (bits/sec)}}$$

Links: physical media

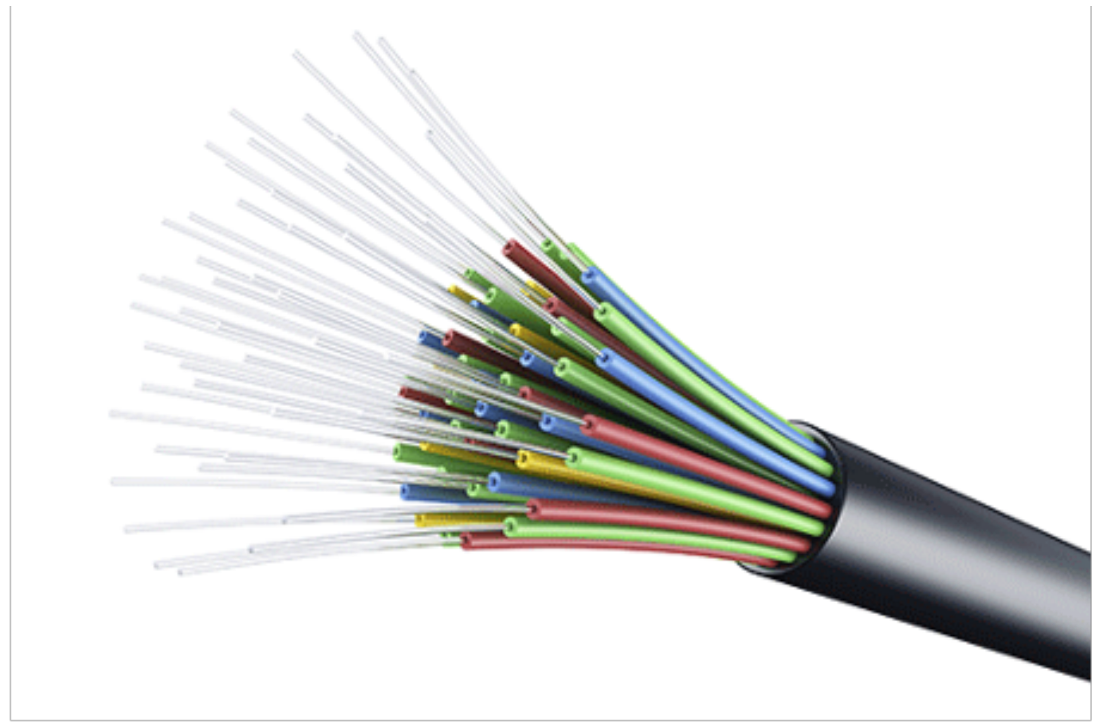
- Bit
  - propagates between transmitter/receiver pairs
- Physical link
  - what lies between transmitter & receiver
- Guided media
  - Signals propagate in solid media: copper, fiber, coax
- Unguided media
  - Signals propagate freely, e.g., radio

Links: physical media

* Twisted pair: two insulated copper wires
  - category 5: 100Mbps, 1Gbps Ethernet
  - category 6: 10Gbps Ethernet



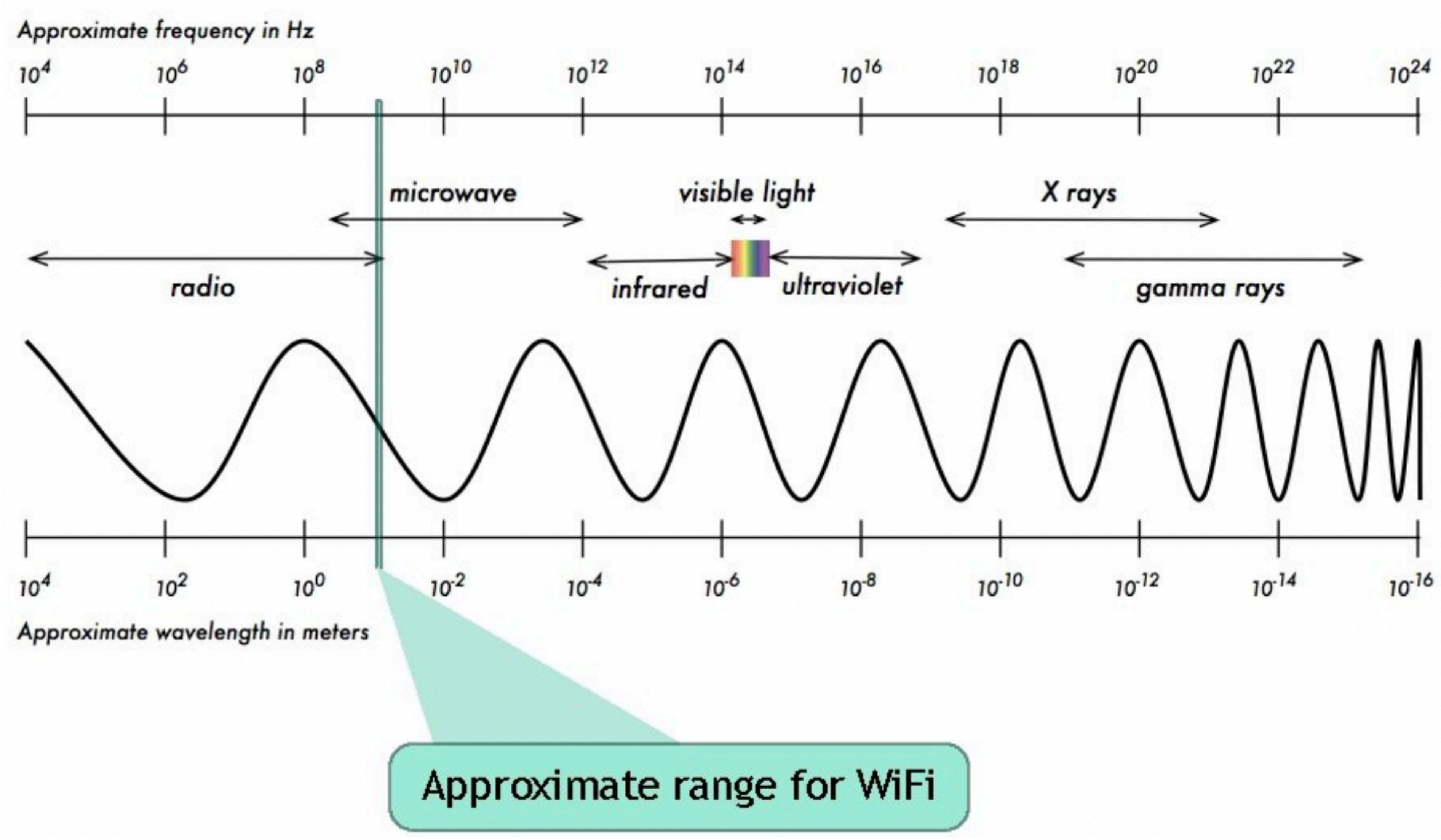
* Fiber optic cable
  - Glass fiber carrying light pulse, each pulse a bit
  - High-speed point-to-point transmission (10's~100's Gbps)
  - Low error rate
    - repeaters spaced far apart
    - Immune to electromagnetic noise



* Coaxial cable
  - Two concentric copper conductors
  - Bidirectional
  - Broadband: multiple frequency channels on cable; 100's Mbps per channel

Links: physical media

* Wireless radio
  - Signal carried in various “bands” in electromagnetic spectrum
  - No physical “wire”
  - Broadcast, “half-duplex” (sender-to-receiver)
  - Propagation environment effects
    - Reflection
    - Obstruction by objects
    - Interference / noise



* Radio link types
  - Wireless LAN (WiFi)
    - 10’s ~ 100’s Mbps; 10's of meters
  - Wide-area (e.g., 4G / 5G cellular)
    - 10’s Mbps (4G) over ~10Km
  - Bluetooth: cable replacement
    - Short distances, limited rates
  - Terrestrial microwave
    - Point-to-point; 45 Mbps channels
  - Satellite
    - Up to < 100 Mbps (Starlink) downlink
    - 270 sec end-to-end delay

Network Core

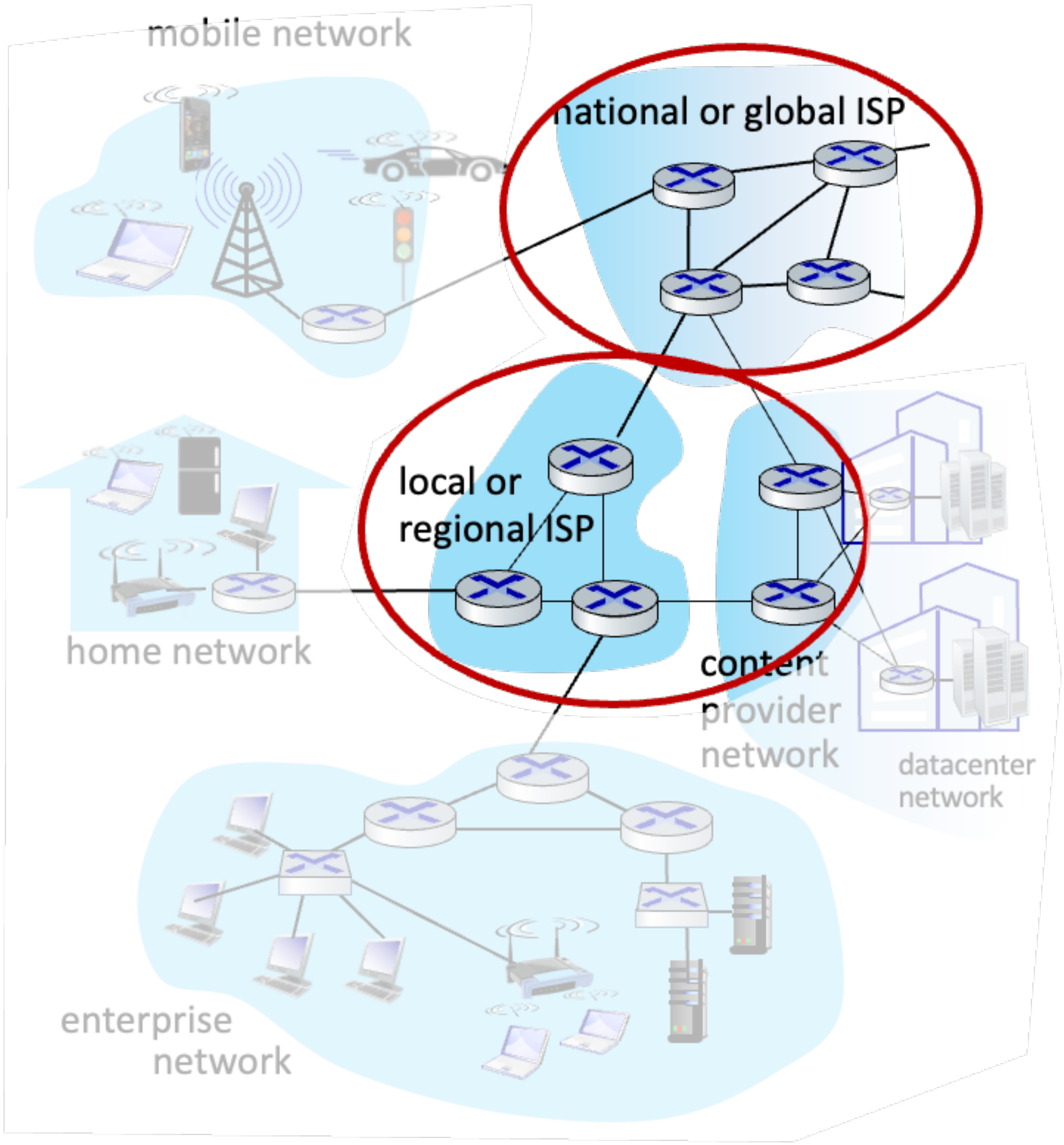
Network core

- Mesh of interconnected routers

- Packet-switching

  - Hosts break application-layer messages into packets

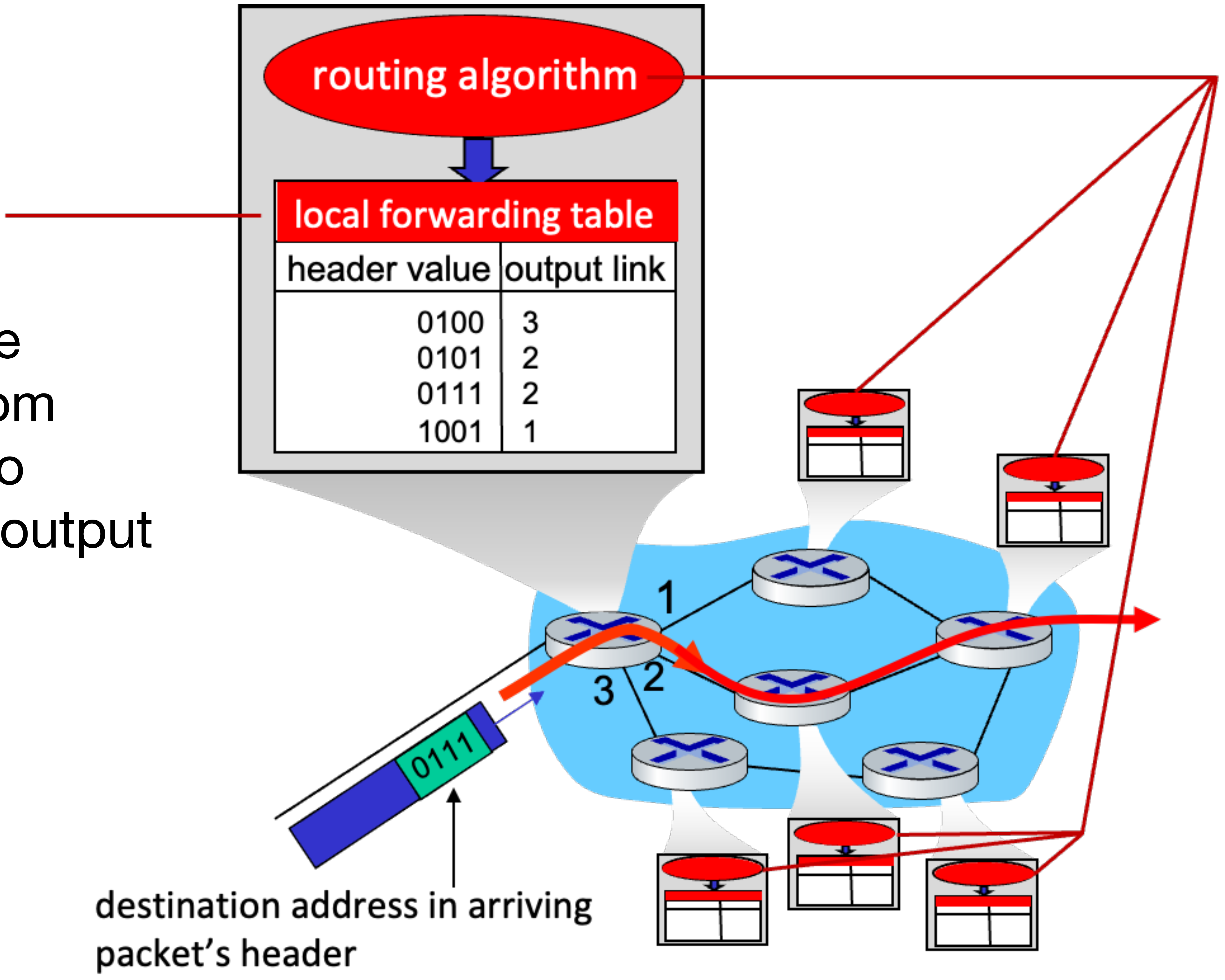
  - Network forwards packets from one router to the next, across links on path from source to destination



Two key network-core functions

Forwarding

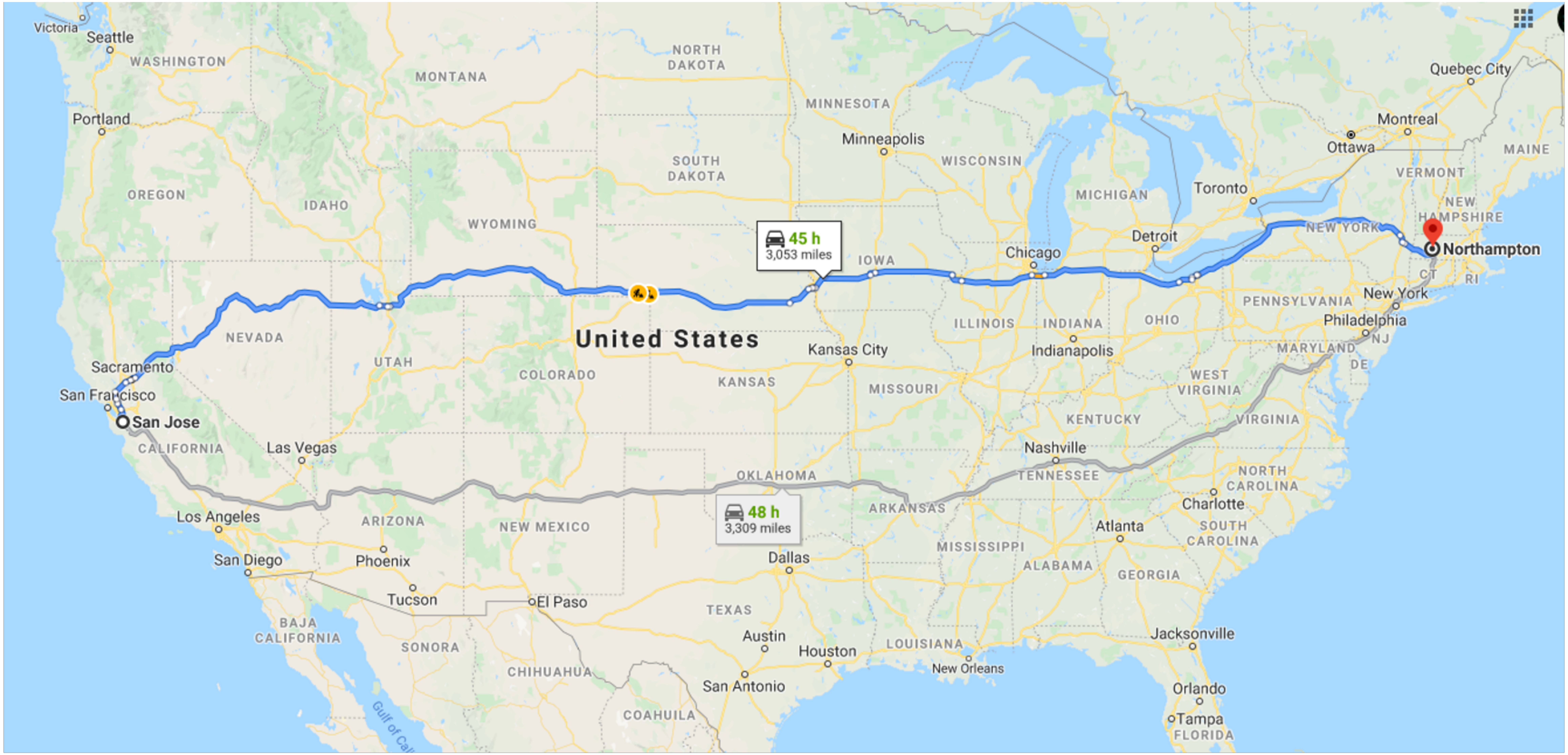
- a.k.a "switching"
- Local action: move arriving packets from router's input link to appropriate router output link



Routing

- Routing Algorithm
- Global action : determine source-destination paths taken by packets

Analogy


45 h
3,053 miles
48 h
3,309 miles
San Jose
Northampton
United States
Victoria
Seattle
WASHINGTON
Portland
OREGON
IDAHO
MONTANA
WYOMING
NORTH DAKOTA
SOUTH DAKOTA
MINNESOTA
Minneapolis
WISCONSIN
MICHIGAN
IOWA
Chicago
Detroit
Toronto
Ottawa
Montreal
Quebec City
MAINE
VERMONT
NEW HAMPSHIRE
NEW YORK
CT
RI
New York
PENNSYLVANIA
Philadelphia
NJ
MARYLAND
DE
OHIO
INDIANA
Indianapolis
ILLINOIS
WEST VIRGINIA
VIRGINIA
KENTUCKY
Nashville
TENNESSEE
NORTH CAROLINA
Charlotte
SOUTH CAROLINA
Atlanta
GEORGIA
ALABAMA
MISSISSIPPI
ARKANSAS
MISSOURI
Kansas City
KANSAS
OKLAHOMA
COLORADO
UTAH
NEVADA
Sacramento
San Francisco
CALIFORNIA
Las Vegas
Los Angeles
San Diego
ARIZONA
Phoenix
Tucson
NEW MEXICO
El Paso
TEXAS
Dallas
Austin
San Antonio
Houston
LOUISIANA
New Orleans
Jacksonville
Orlando
Tampa
FLORIDA
BAJA CALIFORNIA
SONORA
CHIHUAHUA
COAHUILA

Analogy


45 h
3,053 miles
48 h
3,309 miles
routing
United States
San Jose
Northampton

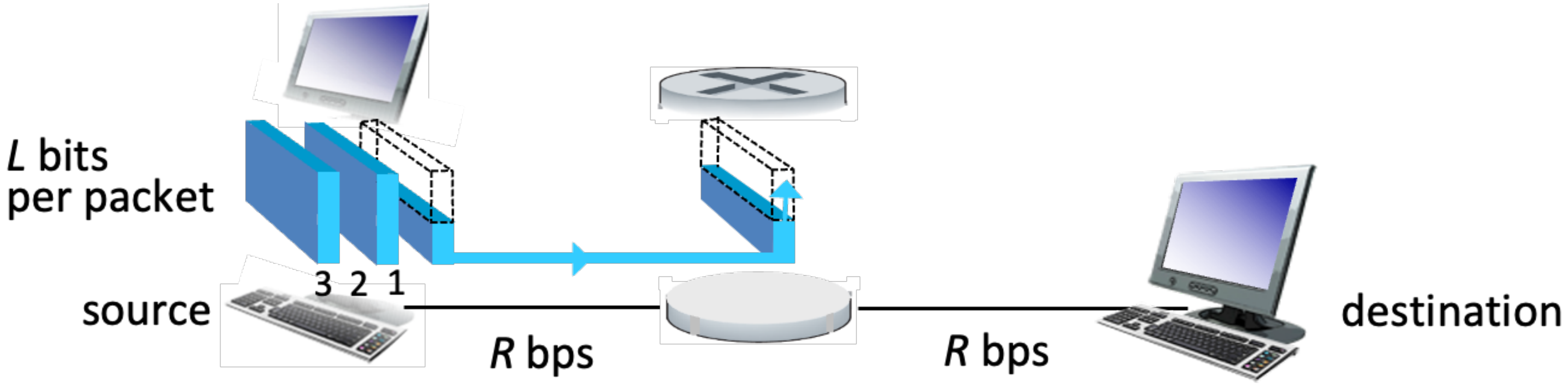
Analogy


forwarding
forwarding
45 h
3,053 miles
48 h
3,309 miles
San Jose
Northampton
United States
Sacramento
Cleveland

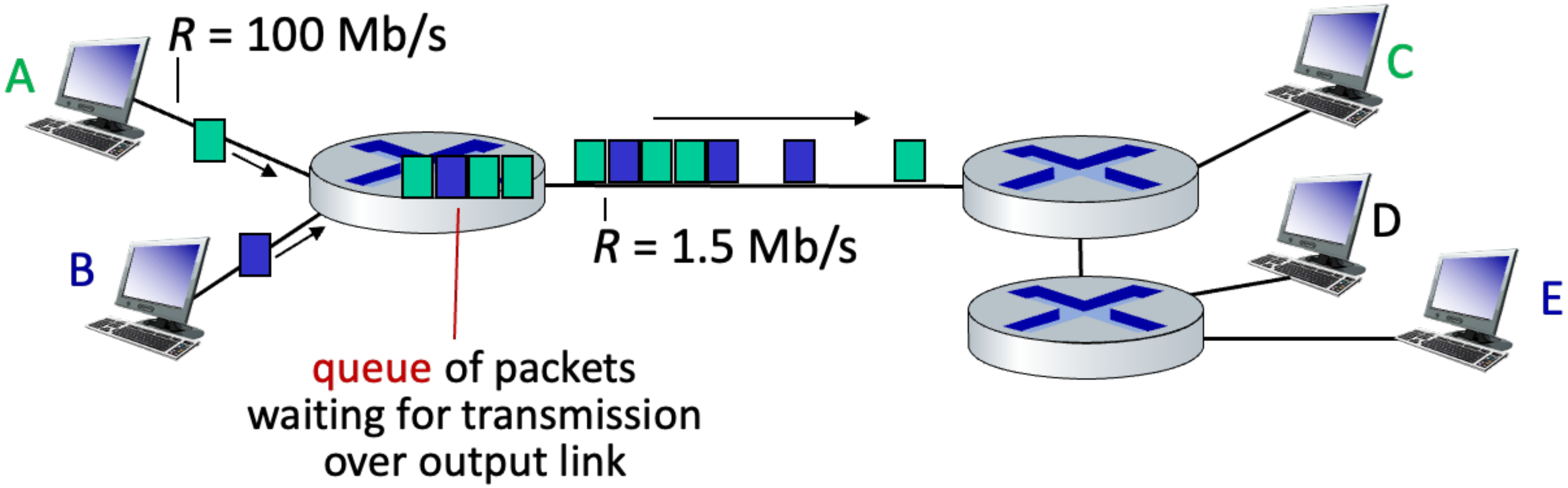
Packet-switching

- Packet transmission delay
  - Takes ***L/R*** seconds to transmit (push out) ***L***-bit packet into link at ***R*** bps
- Store and forward
  - Entire packet must arrive at router before it can be transmitted on next link

One-hop numerical example
- *L* = 10Kbits
- *R* = 100Mbps
- One-hop transmission delay
  = 0.1 msec

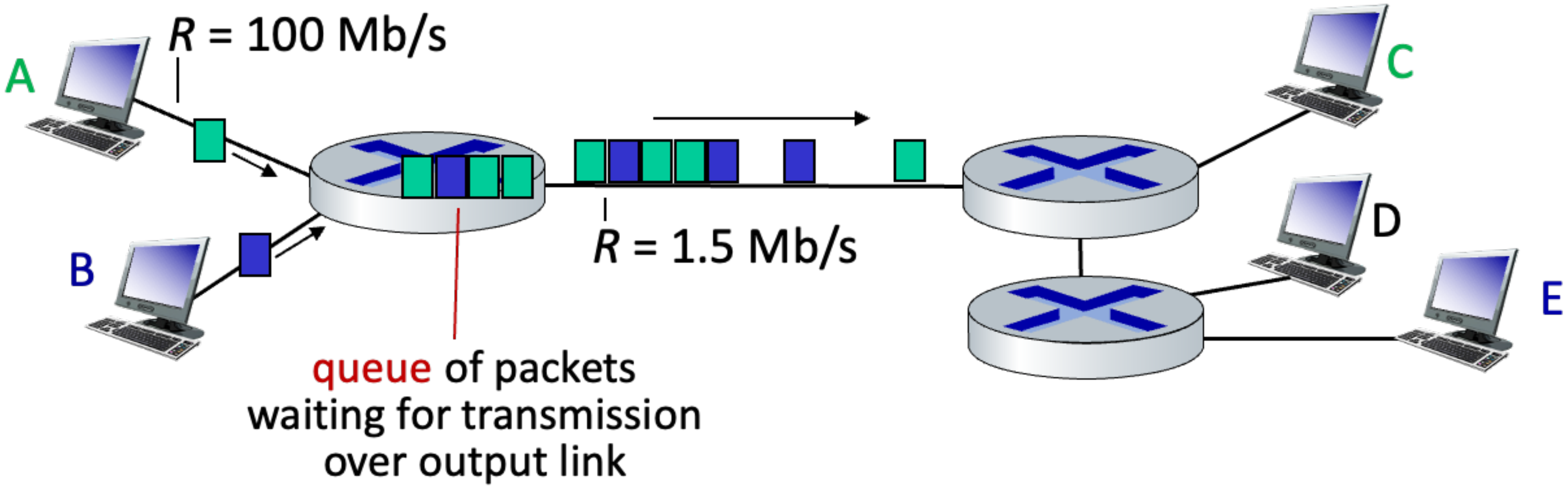


Packet-switching: Queuing



- Queuing occurs when work arrives faster than it can be serviced

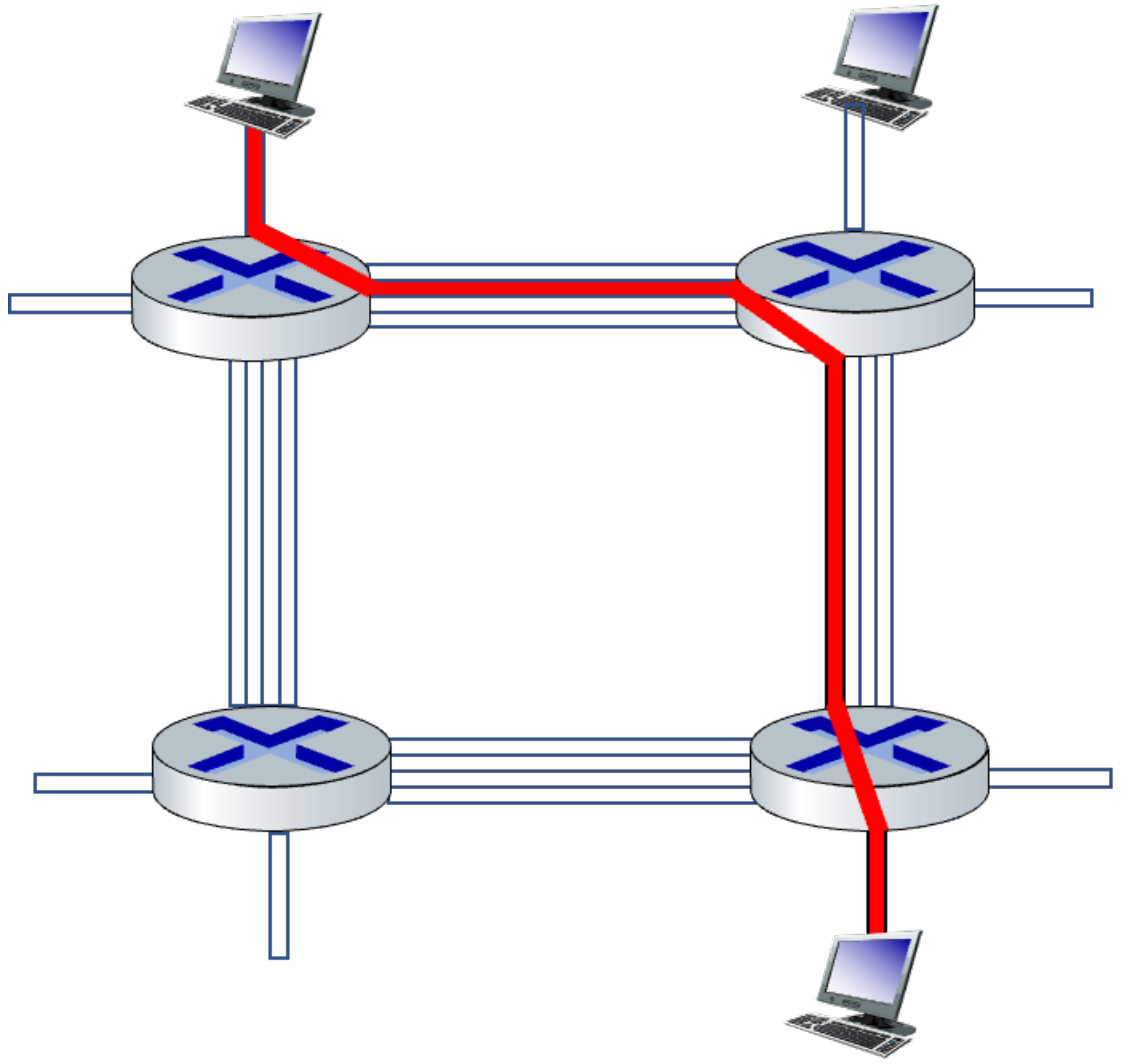
Packet-switching: Queuing



- Packet queuing and loss: if arrival rate (in bps) to link exceeds transmission rate (bps) of link for some period of time
    - Packets will queue, waiting to be transmitted on output link
    - Packets can be dropped (lost) if memory (buffer) in router fills up

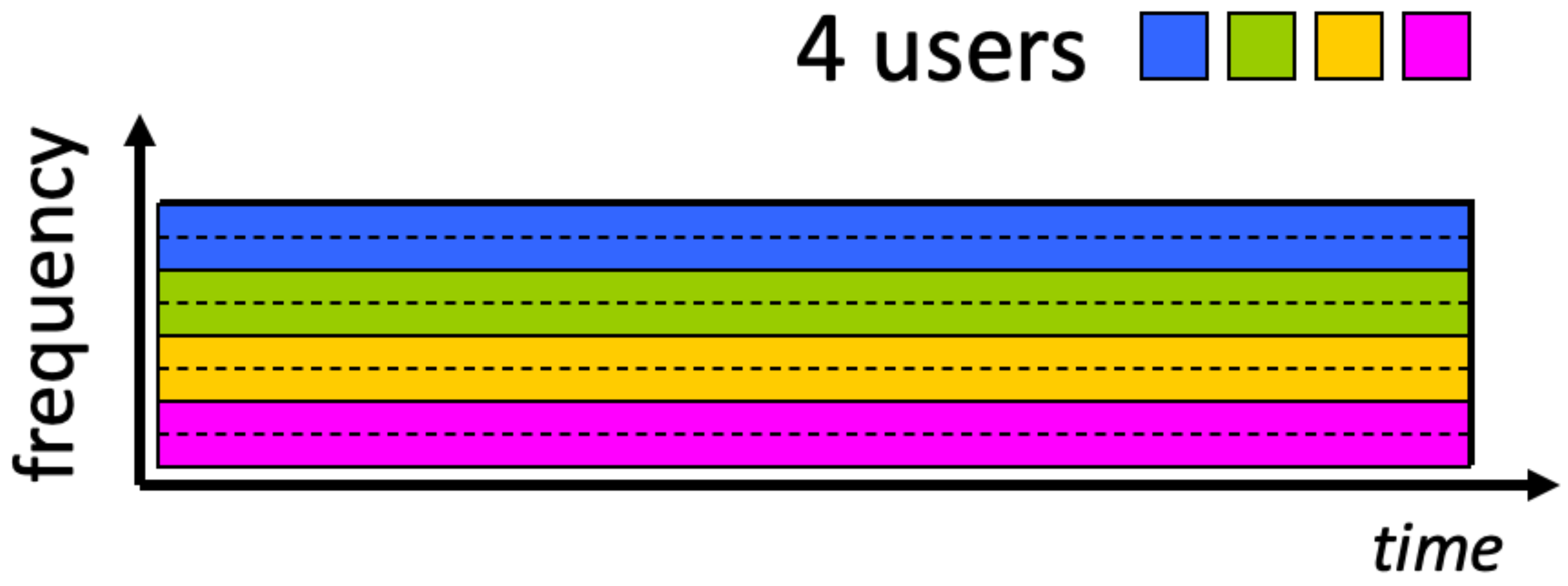
Alternative: Circuit switching

- End-to-end resources allocated to, reserved for call between source and destination
  - Each link has four circuits
    - "call" gets 2^{nd} circuit in top link, and 1^{st} circuit in right link
  - Dedicated resources: no sharing
    - Circuit-like (guaranteed) performance
  - Circuit segment idle if not used by "call" (no sharing)
  - Commonly used in traditional telephone networks

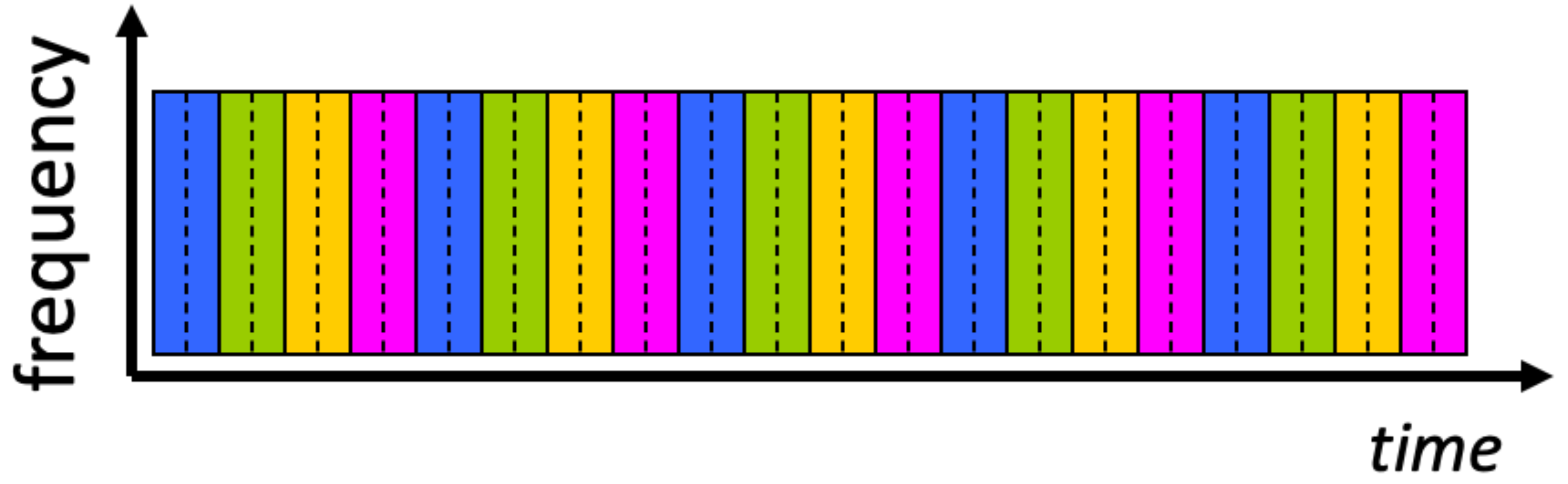


Circuit switching: FDM / TDM

- Frequency Division Multiplexing (FDM)
  - Optical, electromagnetic frequencies divided into (narrow) frequency bands
  - Each call allocated its own band, can transmit at the max rate of that narrow band

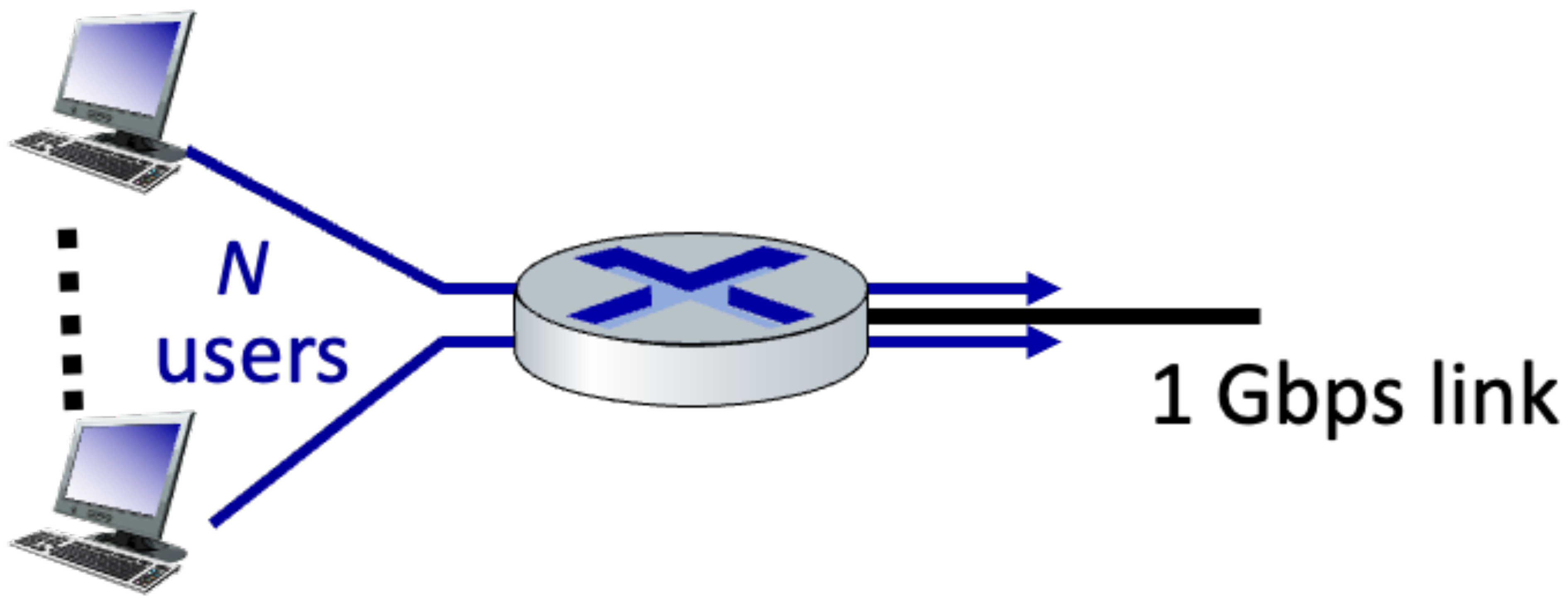


- Time Division Multiplexing (TDM)
  - Time divided into slots
  - Each call allocated periodic slot(s), can transmit at the maximum rate of (wider) frequency band (only) during its time slot(s)



Packet switching vs. Circuit switching

- Example scenario
  - 1Gbps link
  - Each user transmits data at the rate of 100Mbps when “active”
  - “active” during 10% of time



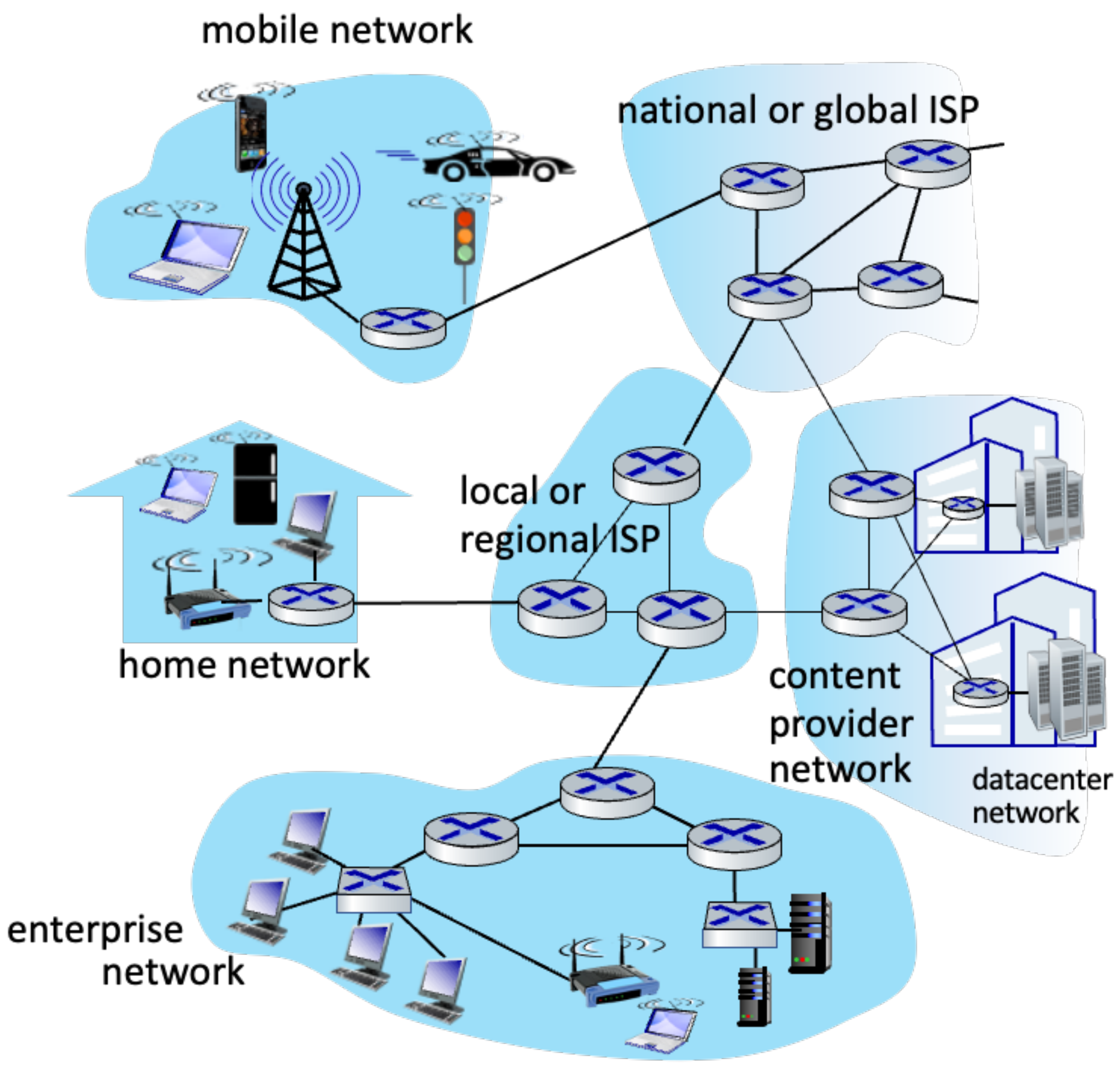
- Question) How many users can use this network under circuit-switching and packet switching?
  - Circuit-switching: 10 users
  - Packet-switching: with 35 users, the probability that more than 10 users become active at the same time is less than 0.0004

Packet switching vs. Circuit switching

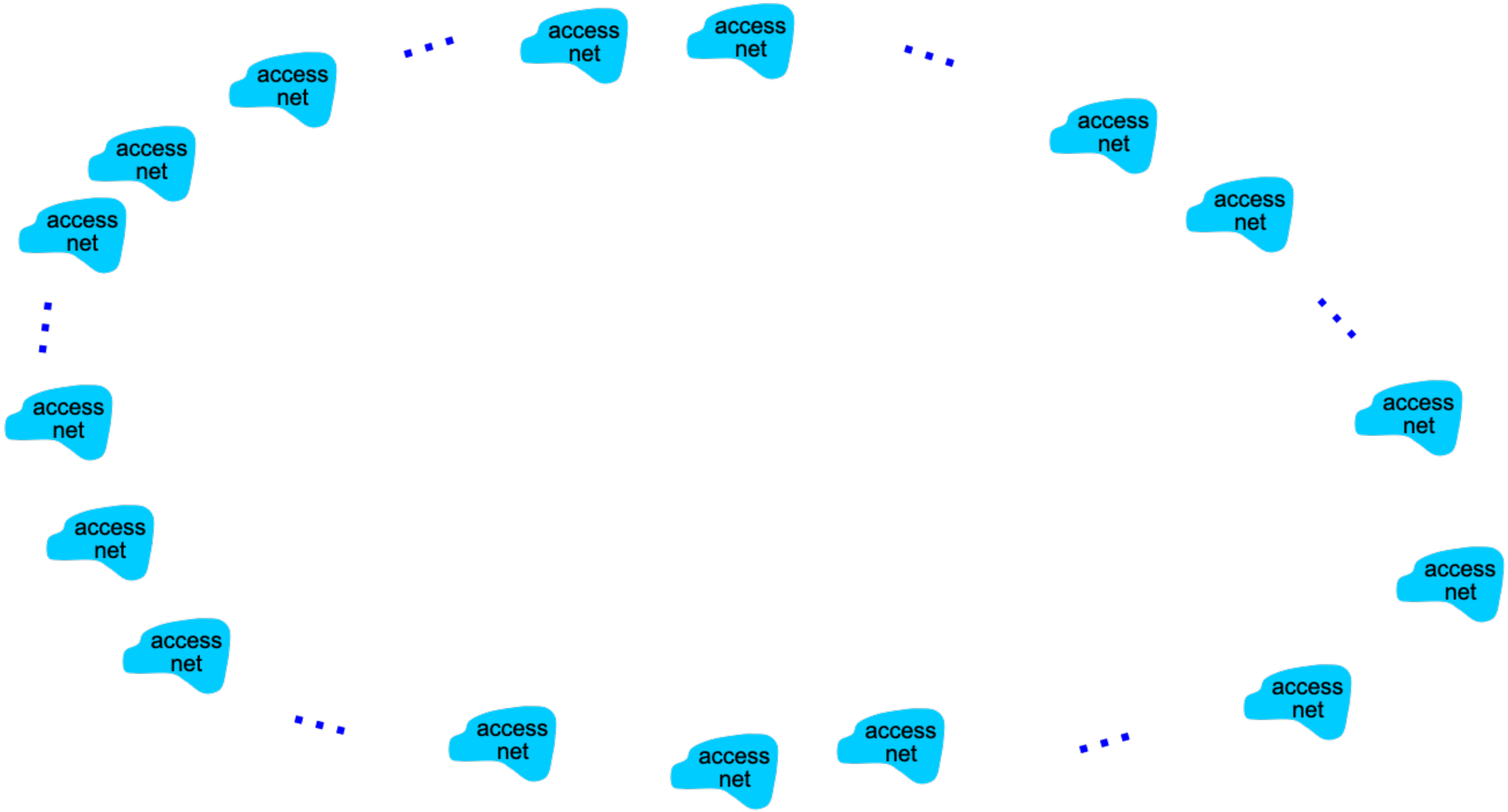
- Is packet switching a “slam dunk winner”?
  - Great for “bursty” data - sometimes has data to send, but at other times not
    - Resource sharing
    - Simpler, no call setup
  - Excessive congestion possible: packet delay and loss due to buffer overflow
    - Protocols needed for reliable data transfer, congestion control
- How to provide circuit-like behavior under packet-switching?
  - It’s complicated. We will discuss various techniques that try to make packet switching as “circuit-like” as possible

Internet structure : a network of networks

- Hosts connect Internet via access Internet Service Provider (ISP)
- Access ISPs in turn must be interconnected
    - Any two hosts (anywhere!) can send packets to each other
- Resulting network of networks is very complex
    - Evolution driven by economics, national policies

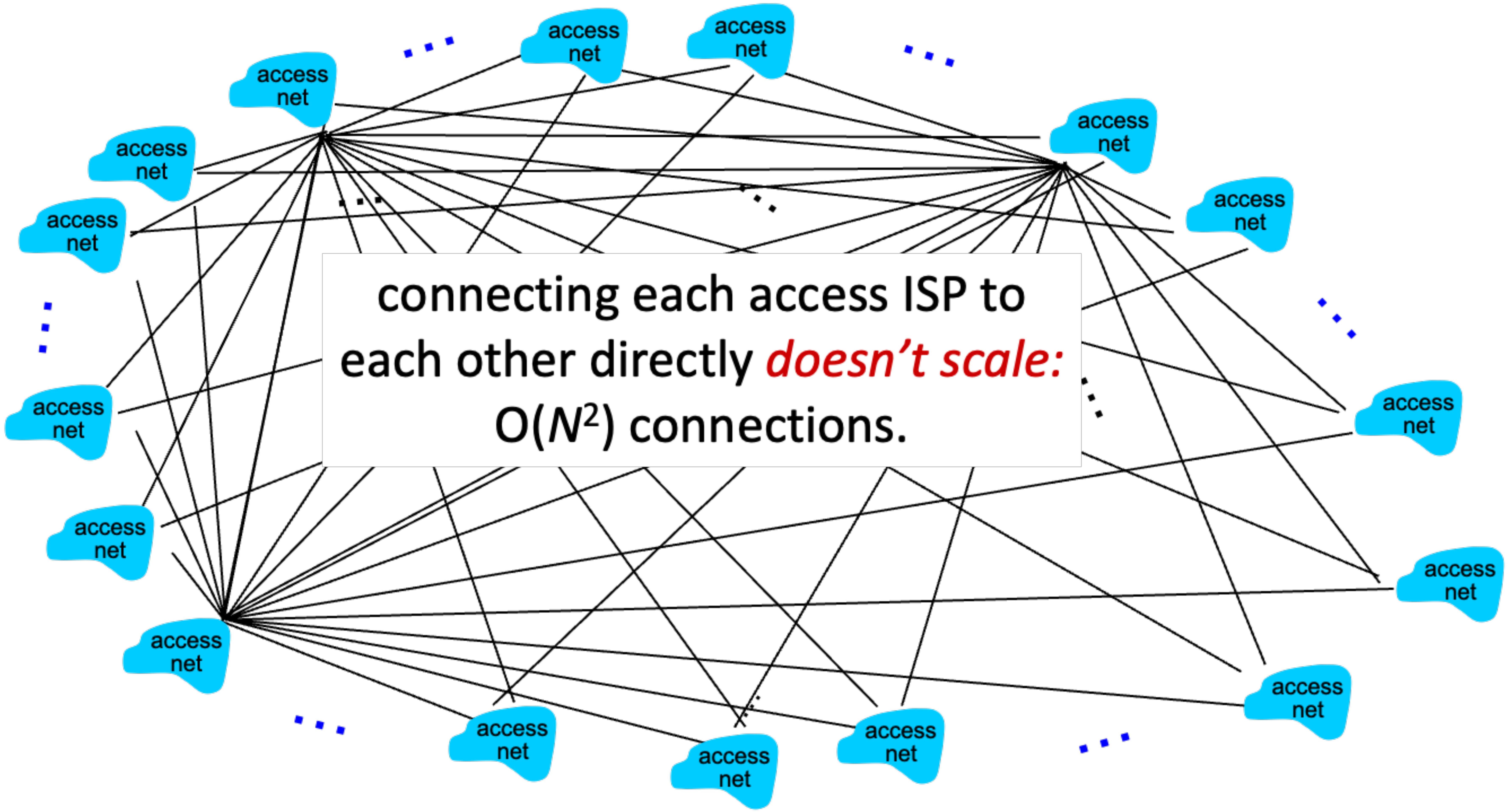


Internet structure : a network of networks



Q) Given millions of access ISPs, how to connect them together?

Internet structure : a network of networks



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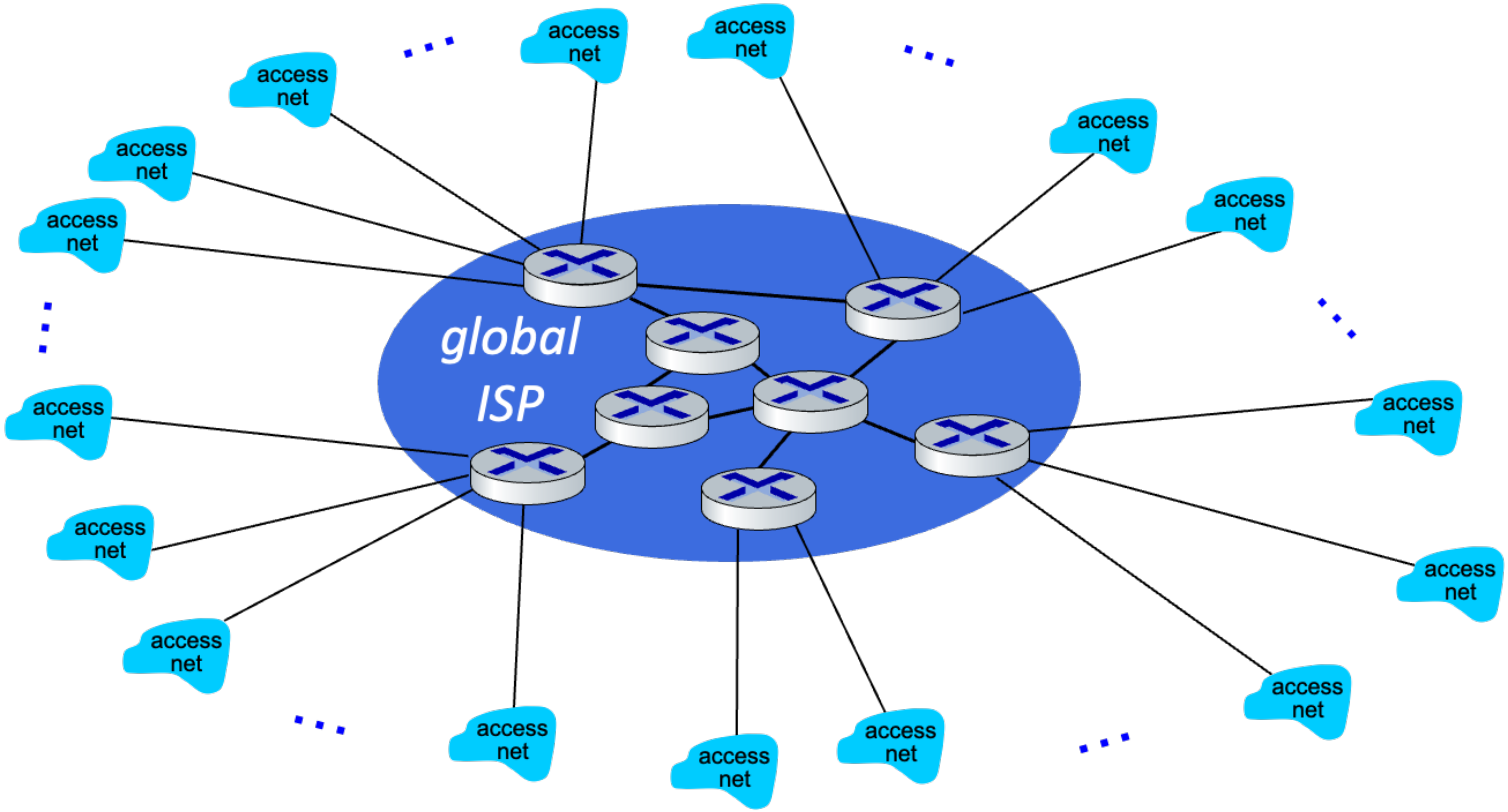
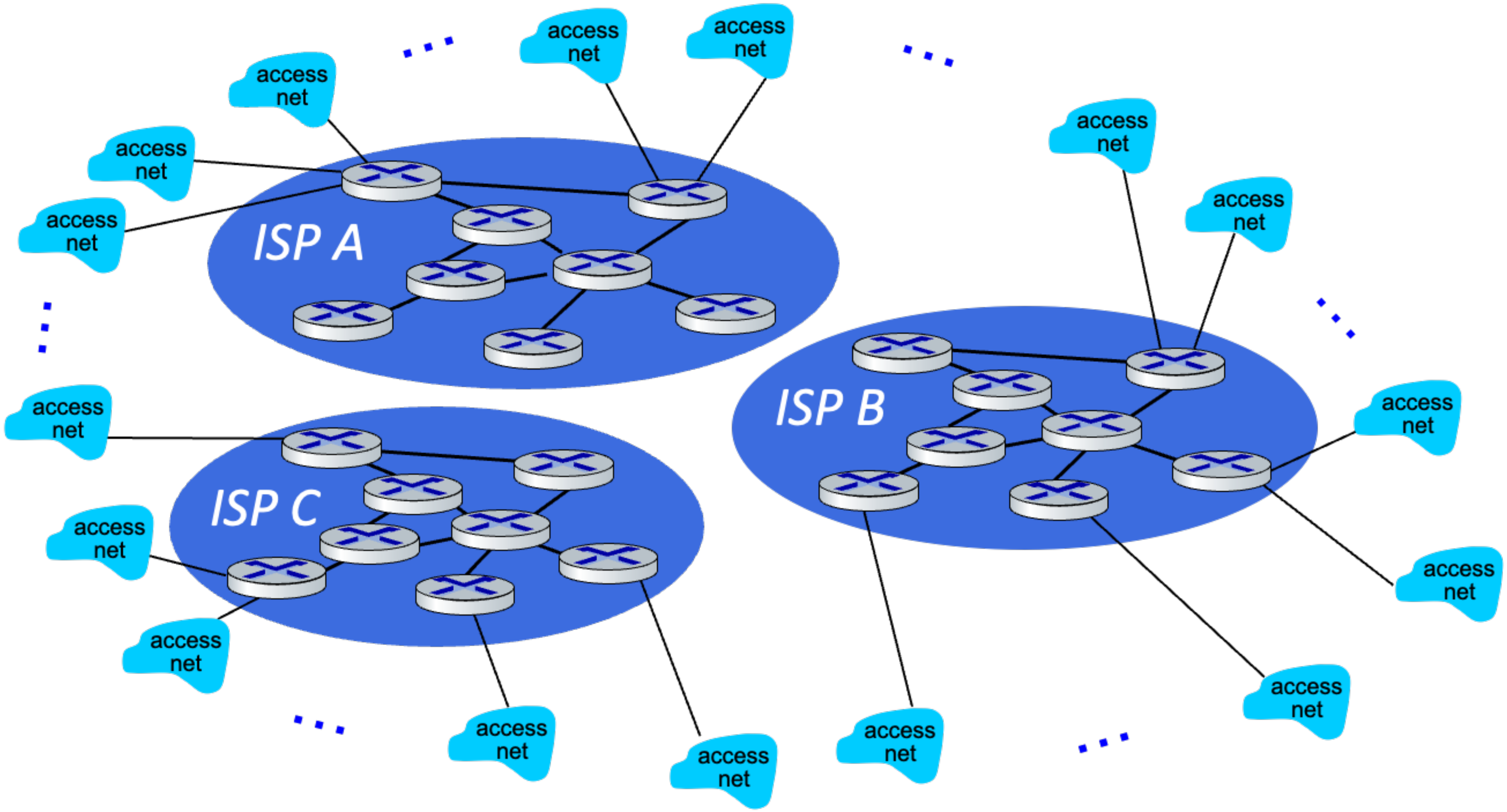
Internet structure : a network of networks



Opt.1) Connect each access ISP to one global transit ISP?

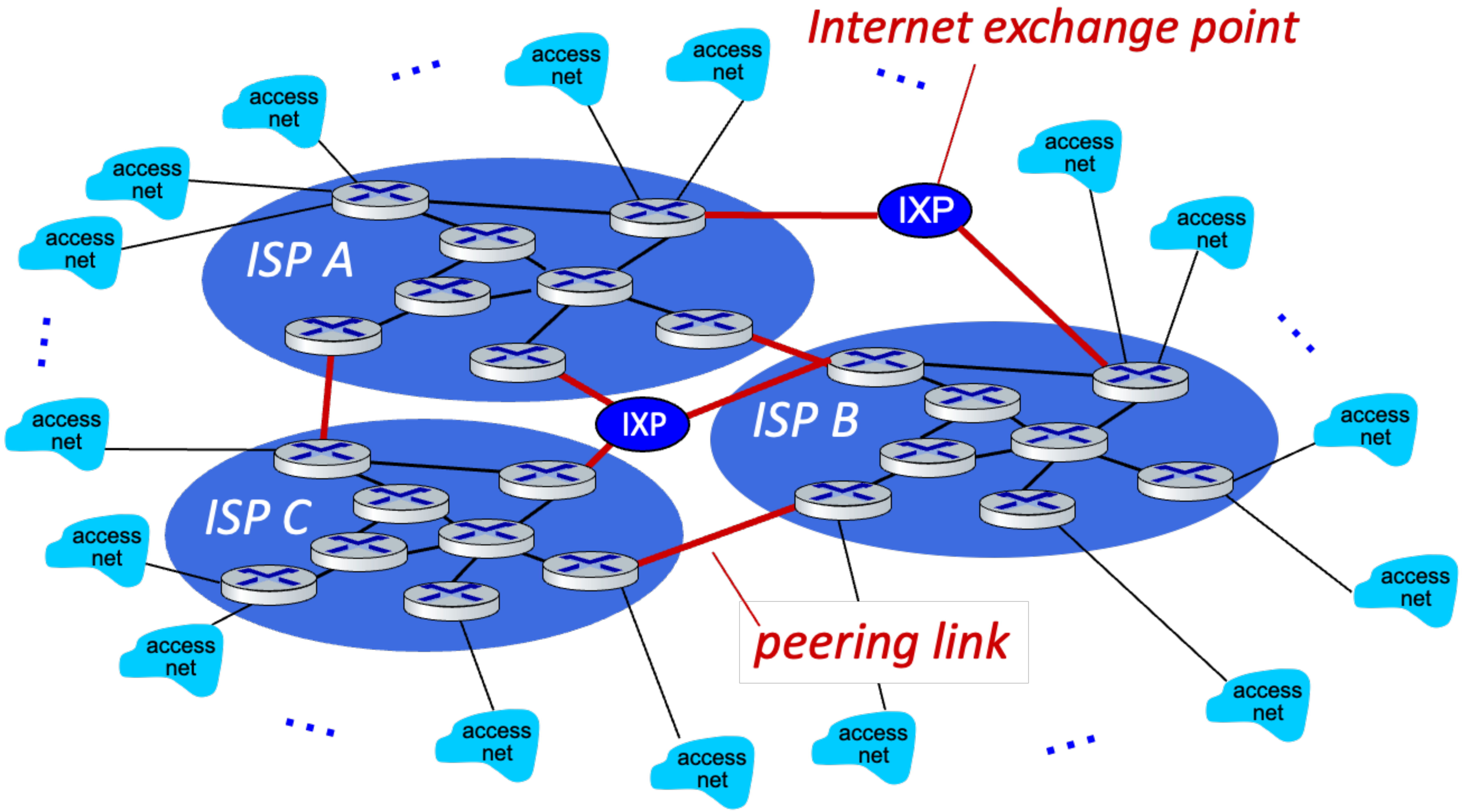
→ Customer and Provider ISPs have economic agreement

Internet structure : a network of networks



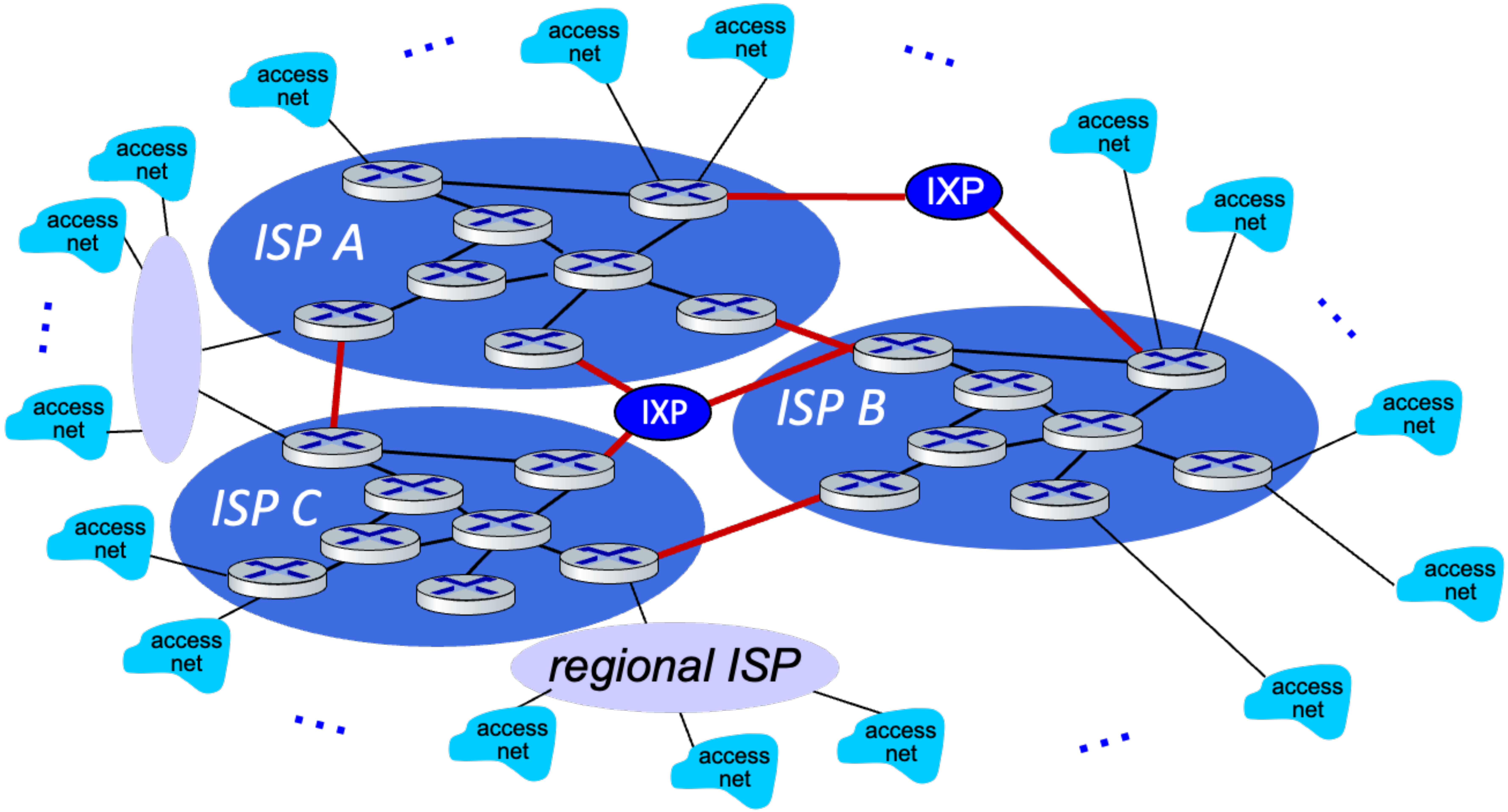
If one global ISP is viable business, there will be competitors …

Internet structure : a network of networks



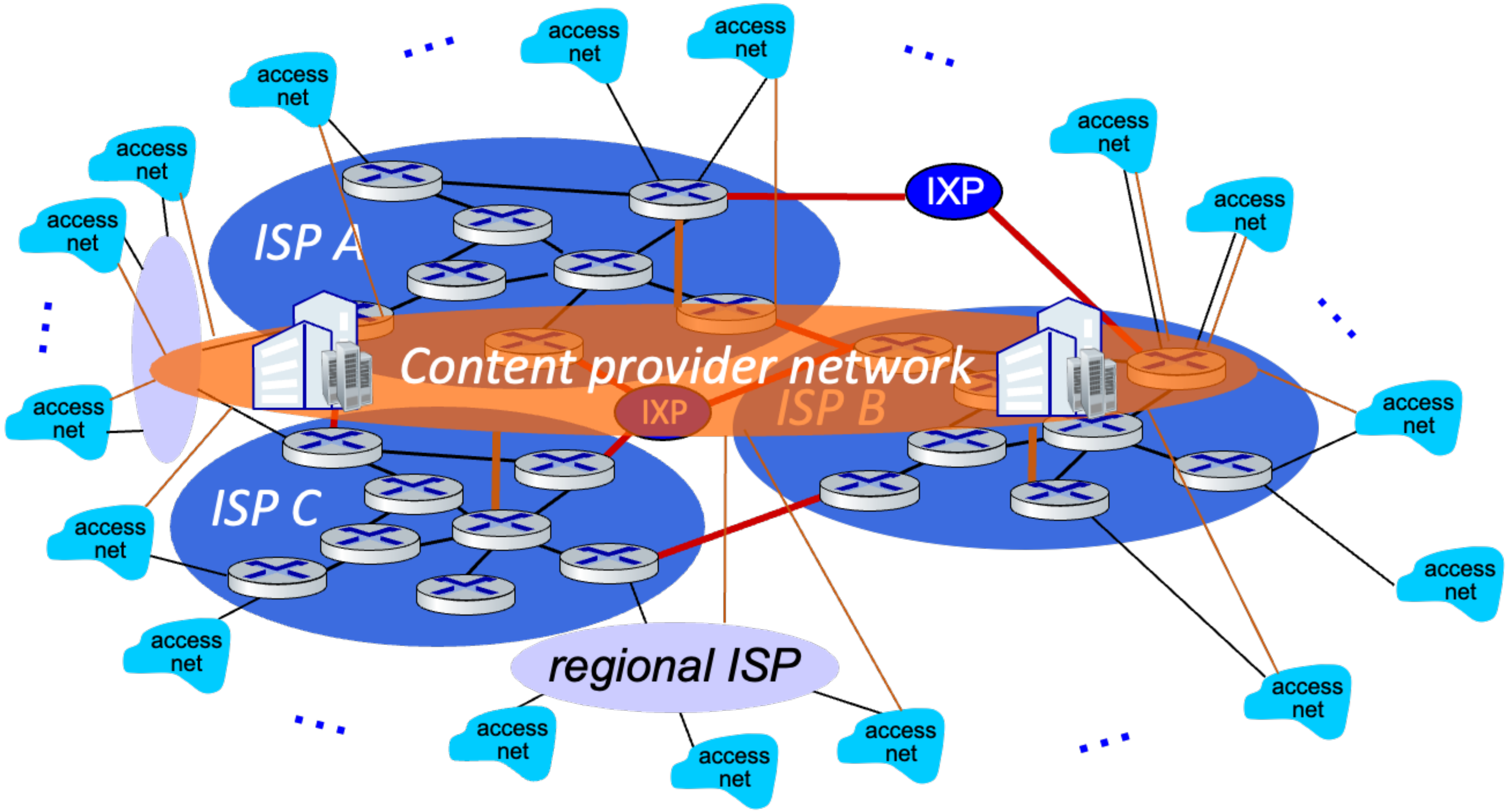
If one global ISP is viable business, there will be competitors …who will want to be connected

Internet structure : a network of networks



… and regional networks may arise to connect access nets to ISPs

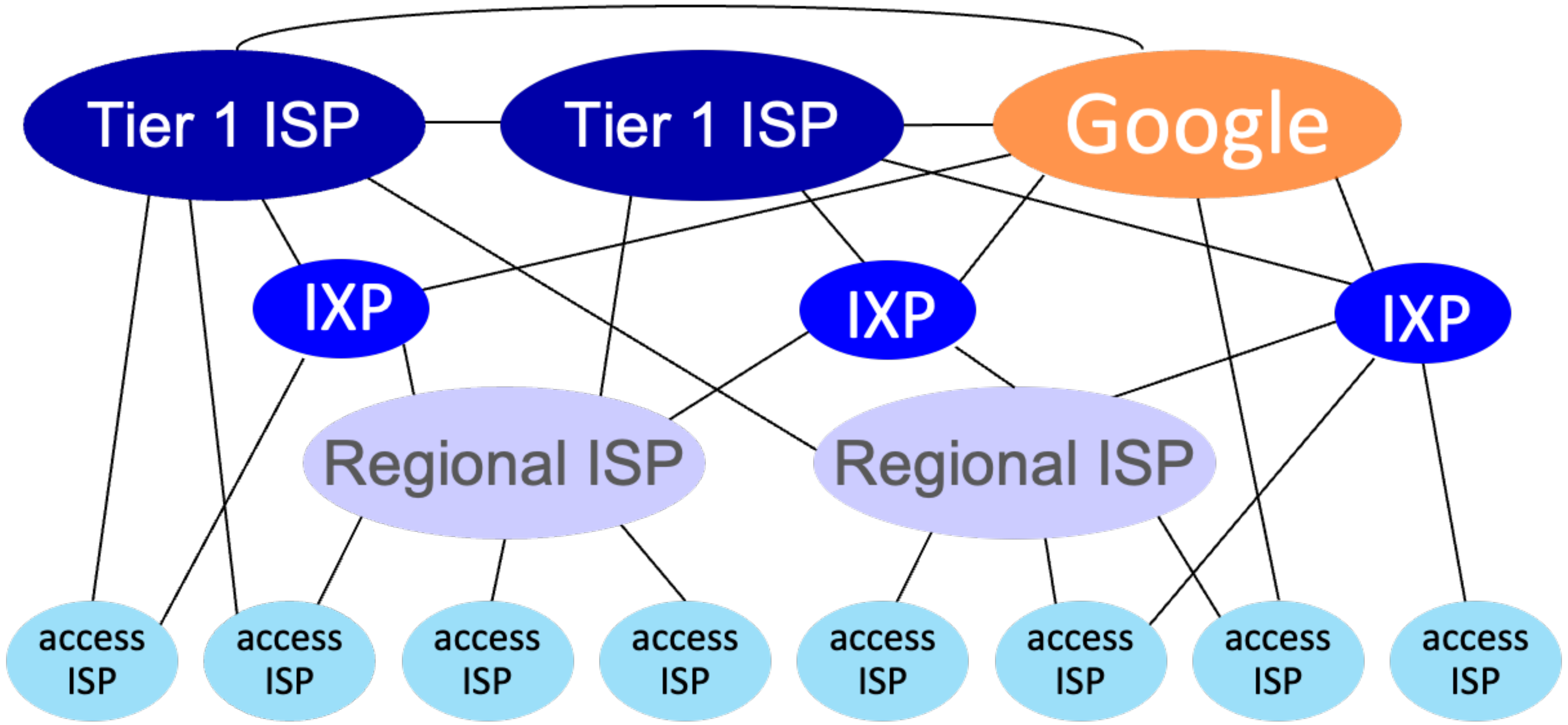
Internet structure : a network of networks



… and content provider networks (e.g., Google, Microsoft, Akamai) may run their own network, to bring services, content closer to end users

Internet structure : a network of networks

- At “center”, small number of well-connected large networks
  - “tier-1” commercial ISPs (e.g., Level 3, Sprint, AT&T, NTT), national & international coverage
  - Content provider networks (e.g., Google, Meta): private network that connects its data center to Internet, often bypassing “tier-1” regional ISPs

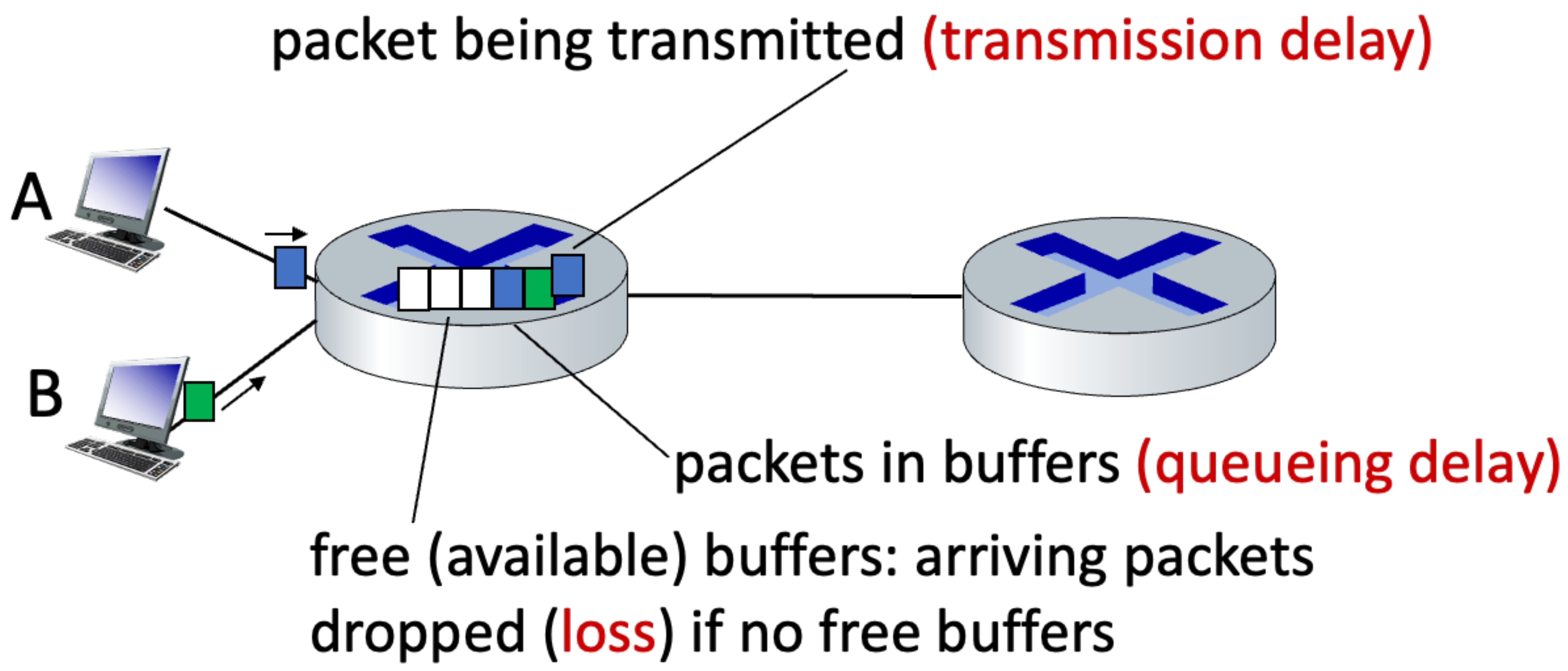


Performance: Loss, Delay, Throughput

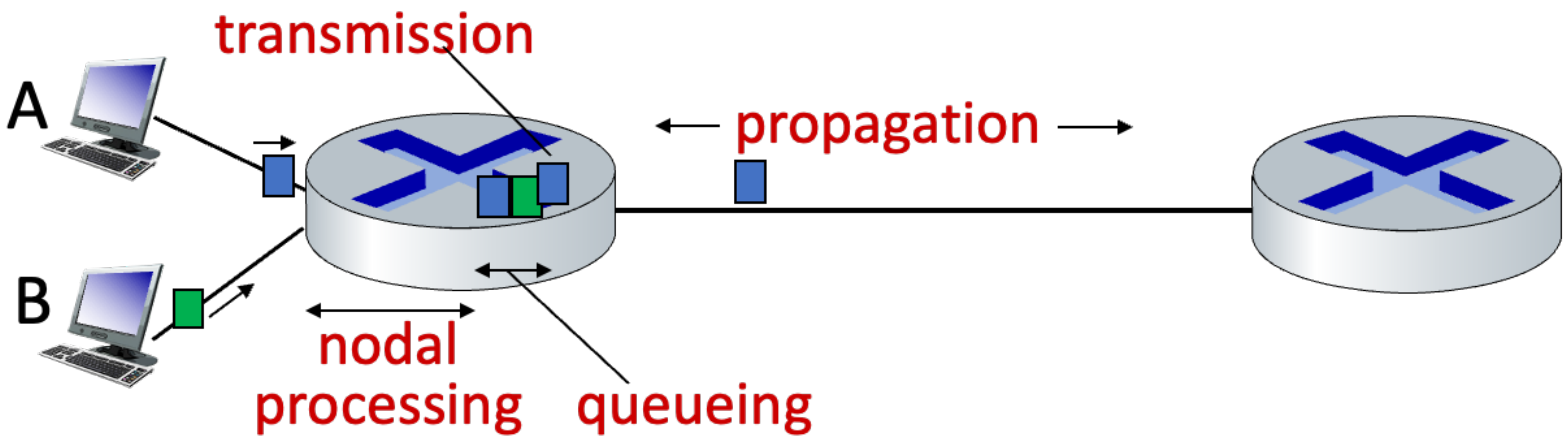
How do packet Delay and Loss occur?

- Packets queue in router buffers, waiting for turn for transmission
  - Queue length grows when arrival rate to link (temporarily) exceeds output link capacity

- Packet loss occurs when memory to hold queued packets fills up



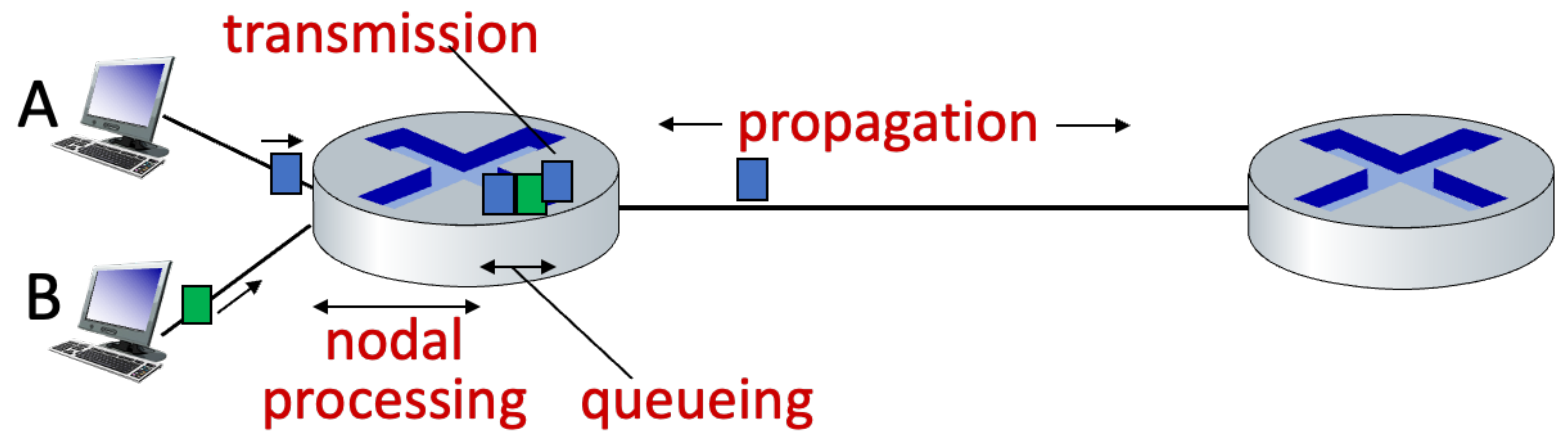
Packet delay



$$d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$$

- d_{proc} : nodal processing
  - Check bit errors
  - Determine output link
  - Typically < microseconds
- d_{queue} : queuing delay
  - Time waiting at output link for transmission
  - Depends on congestion level of router

Packet delay

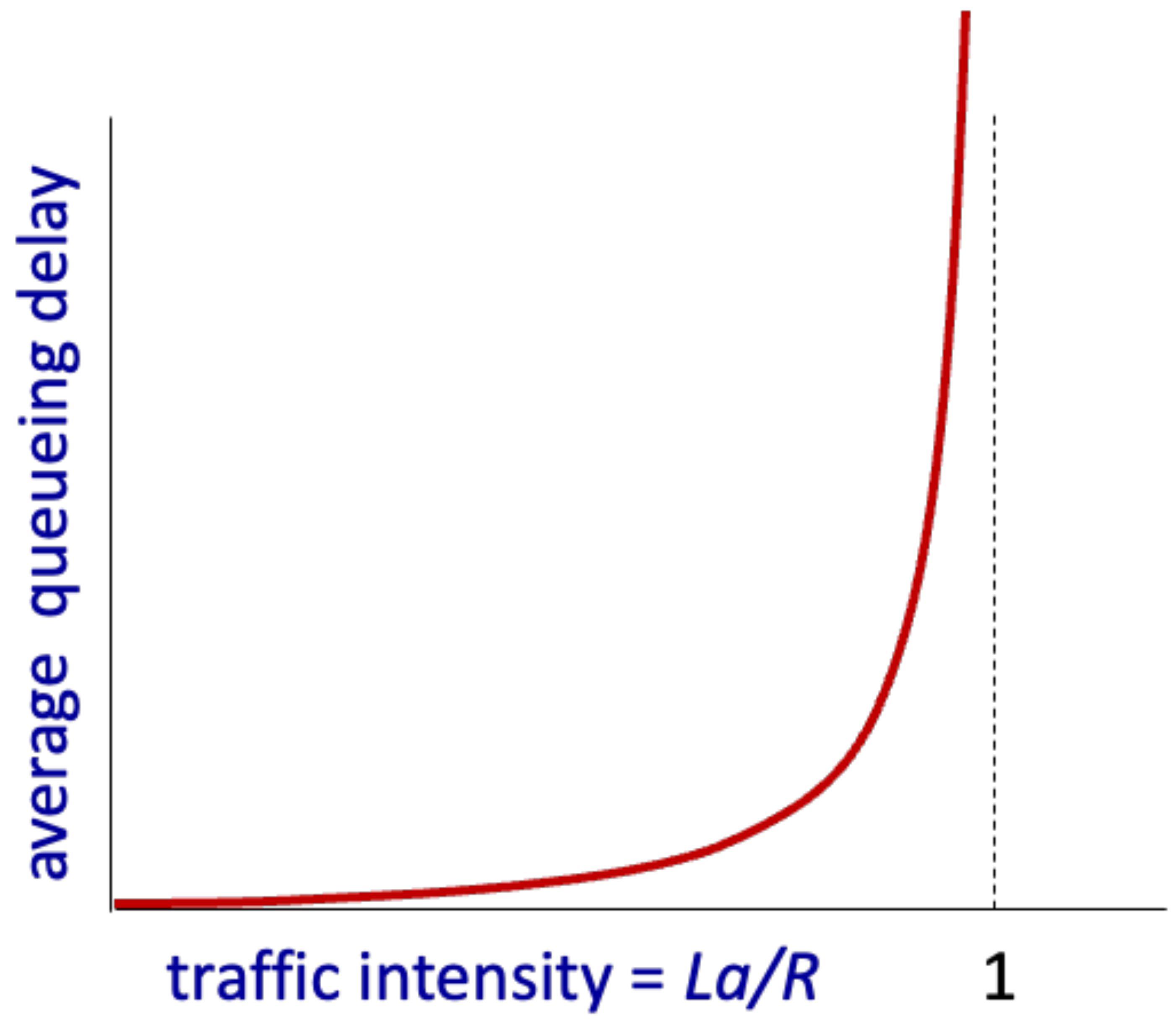


$$d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$$

- d_{trans} : transmission delay
  - L: packet length (bits)
  - R: link transmission rate
  - $d_{trans} = L / R$
- d_{prop} : propagation delay
  - d: length of physical link
  - s: propagation speed (~2×10^8m/s)
  - $d_{prop} = d / s$

Packet queuing delay

- a: average arrival rate
- L: packet length (bits)
- R: link bandwidth (bit transmission rate)

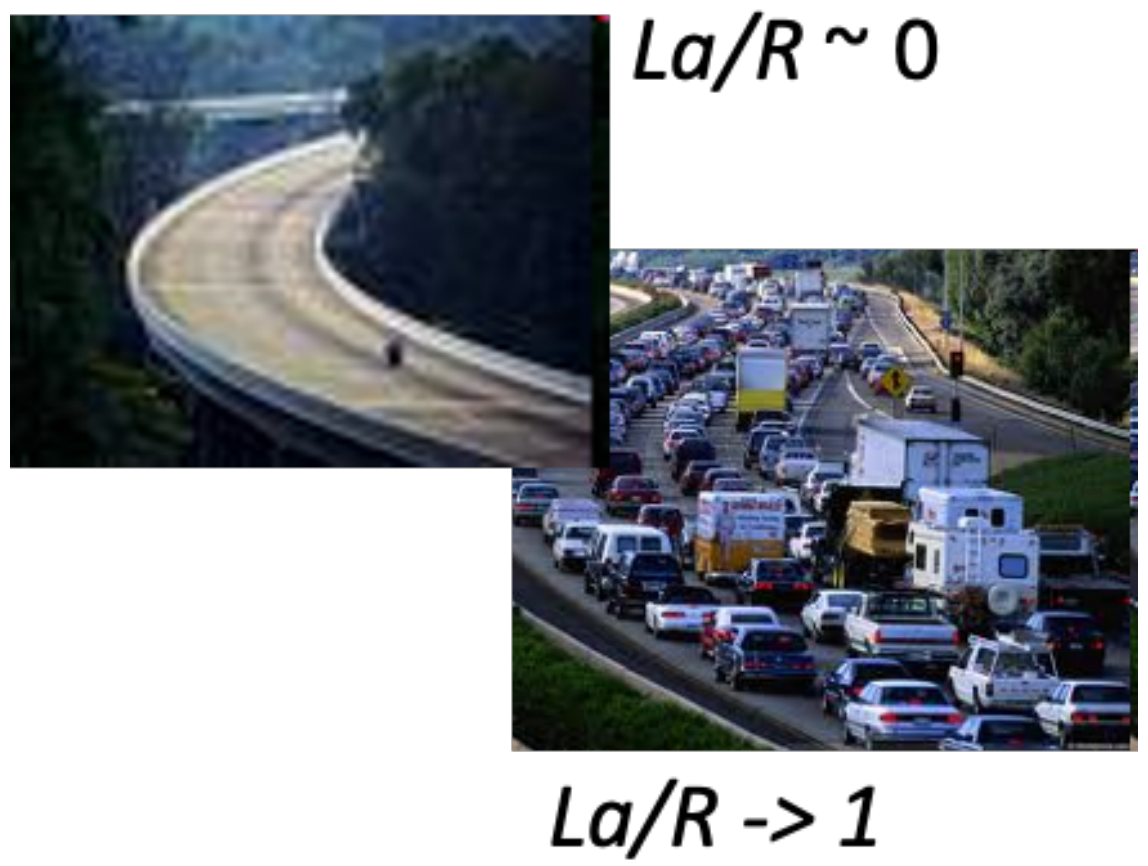


$$\boxed{\text{Traffic intensity} = \frac{La}{R} : \frac{\textbf{arrival rate of bits}}{\textbf{service rate of bits}}}$$

$\frac{La}{R}$ ~ 0 : average queuing delay small

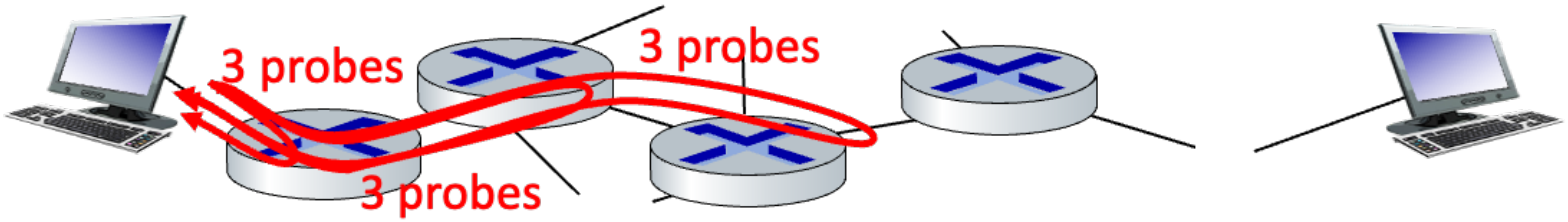
$\frac{La}{R} \rightarrow 1$: average queuing delay large

$\frac{La}{R} > 1$: "work" is arriving more than can be serviced: average delay infinity



Real Internet delay and routes

- What do real Internet delay & route look like?
- "traceroute" program provides delay measurement from source to router along end-to-end Internet path towards destination
  - Send three packets that will reach router i on path towards destination
  - Router i will return packets to sender
  - Sender measures time interval between transmission and reply
  - Increment i and repeat the process



Real Internet delay and routes

traceroute: gaia.cs.umass.edu to www.eurecom.fr

3 delay measurements from
gaia.cs.umass.edu to cs-gw.cs.umass.edu

```
1  cs-gw (128.119.240.254)  1 ms  1 ms  2 ms
2  border1-rt-fa5-1-0.gw.umass.edu (128.119.3.145)  1 ms  1 ms  2 ms
3  cht-vbns.gw.umass.edu (128.119.3.130)  6 ms 5 ms 5 ms
4  jn1-at1-0-0-19.wor.vbns.net (204.147.132.129)  16 ms 11 ms 13 ms
5  jn1-so7-0-0-0.wae.vbns.net (204.147.136.136)  21 ms 18 ms 18 ms
6  abilene-vbns.abilene.ucaid.edu (198.32.11.9)  22 ms  18 ms  22 ms
7  nycm-wash.abilene.ucaid.edu (198.32.8.46)  22 ms  22 ms  22 ms
8  62.40.103.253 (62.40.103.253)  104 ms 109 ms 106 ms
9  de2-1.de1.de.geant.net (62.40.96.129)  109 ms 102 ms 104 ms
10  de.fr1.fr.geant.net (62.40.96.50)  113 ms 121 ms 114 ms
11  renater-gw.fr1.fr.geant.net (62.40.103.54)  112 ms  114 ms  112 ms
12  nio-n2.cssi.renater.fr (193.51.206.13)  111 ms  114 ms  116 ms
13  nice.cssi.renater.fr (195.220.98.102)  123 ms  125 ms  124 ms
14  r3t2-nice.cssi.renater.fr (195.220.98.110)  126 ms  126 ms  124 ms
15  eurecom-valbonne.r3t2.ft.net (193.48.50.54)  135 ms  128 ms  133 ms
16  194.214.211.25 (194.214.211.25)  126 ms  128 ms  126 ms
17  * * *
18  * * *
19  fantasia.eurecom.fr (193.55.113.142)  132 ms  128 ms  136 ms
```

trans-oceanic link

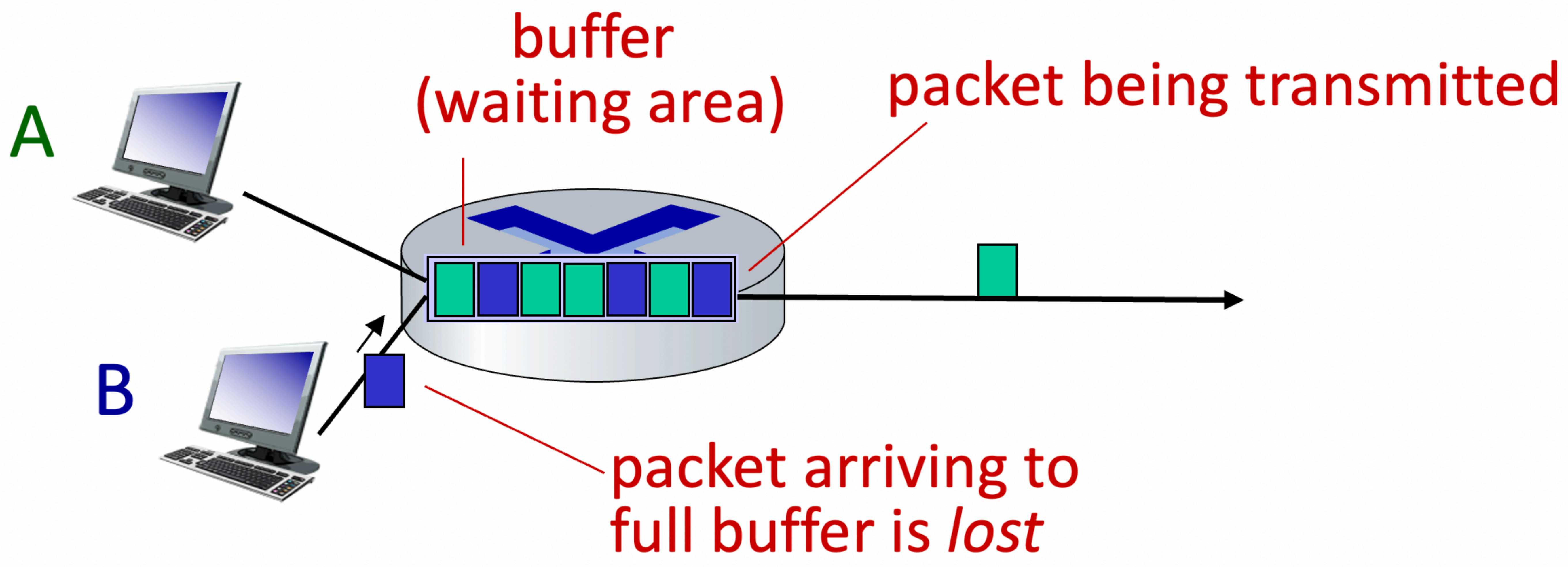
looks like delays *decrease*! Why?

* means no response (probe lost, router not replying)

* Do some traceroutes from exotic countries at www.traceroute.org

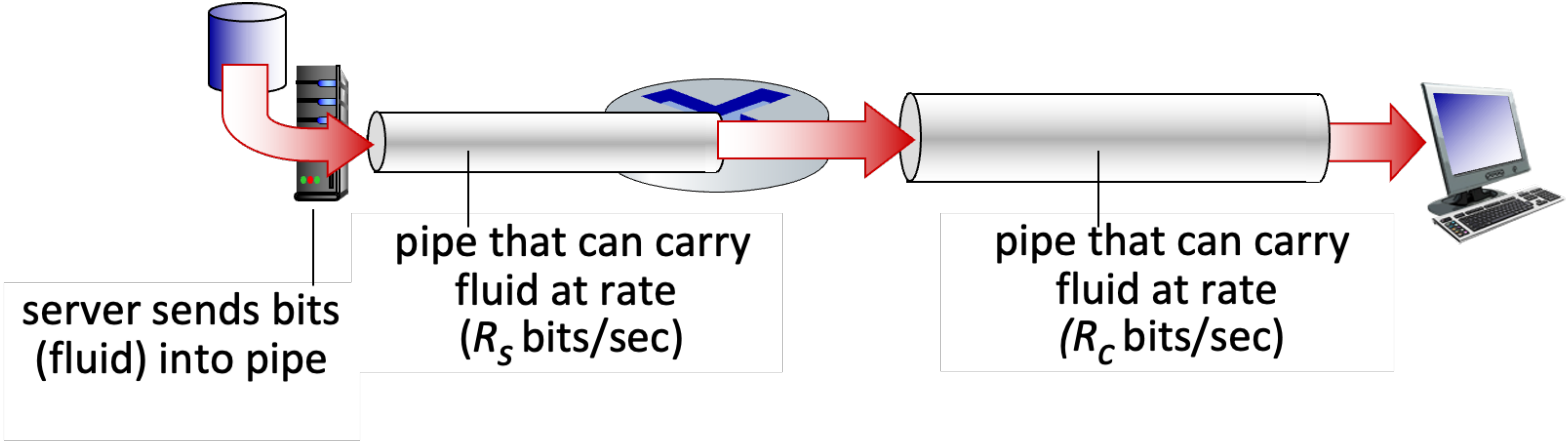
Packet loss

- Queue (a.k.a buffer) has finite capacity
- Packets arriving to full queue will be dropped (a.k.a. lost)
- Lost packets may be retransmitted by previous node, by source end system or not at all



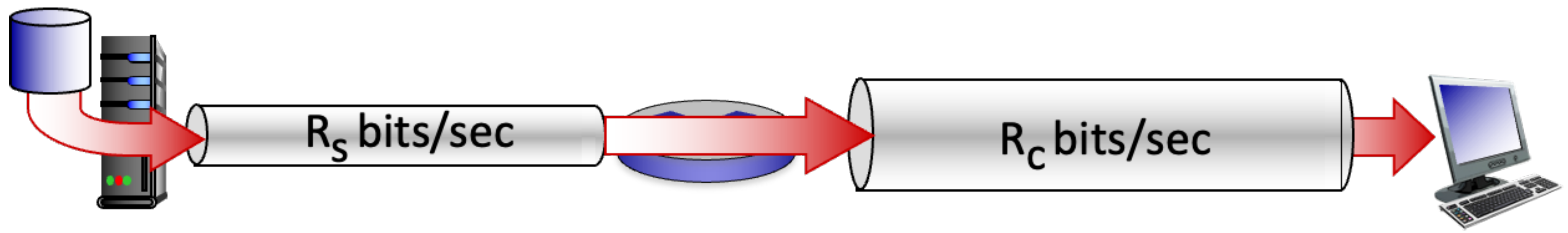
Throughput

- Throughput (bits/time unit)
  - Rate at which bits are being sent from sender to receiver
  - Instantaneous throughput : rate at given point in time
  - Average throughput : rate over longer period of time

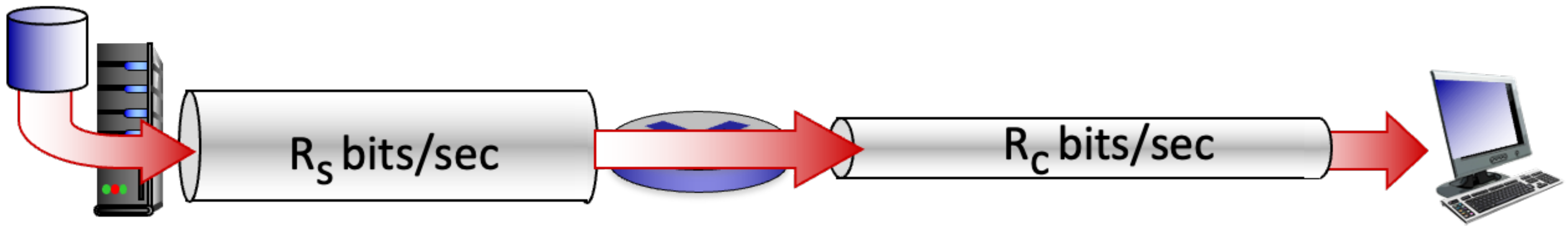


Throughput

- $R_s < R_c$: what is the average end-to-end throughput?

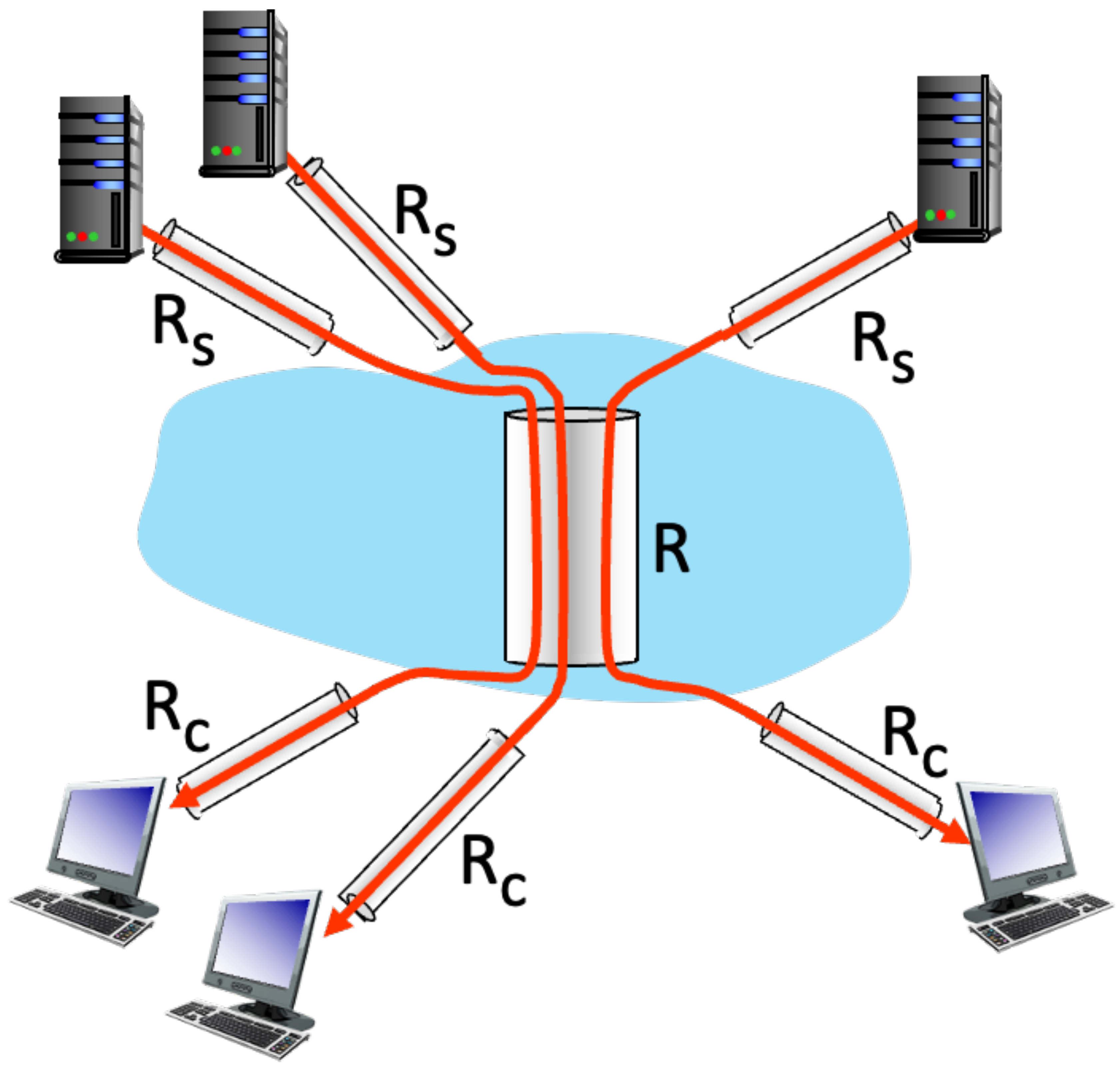


- $R_s < R_c$: what is the average end-to-end throughput?



Bottleneck link
⇒ link on the end-to-end path that determines the end-to-end throughput

Throughput : network scenario



10 connections (fairly) share backbone bottleneck link *R* bits/sec

- Per-connection end-to-end throughput
  - min (R_c, R_s, $R/10$)
- In practice
  - R_c or R_s is often bottleneck

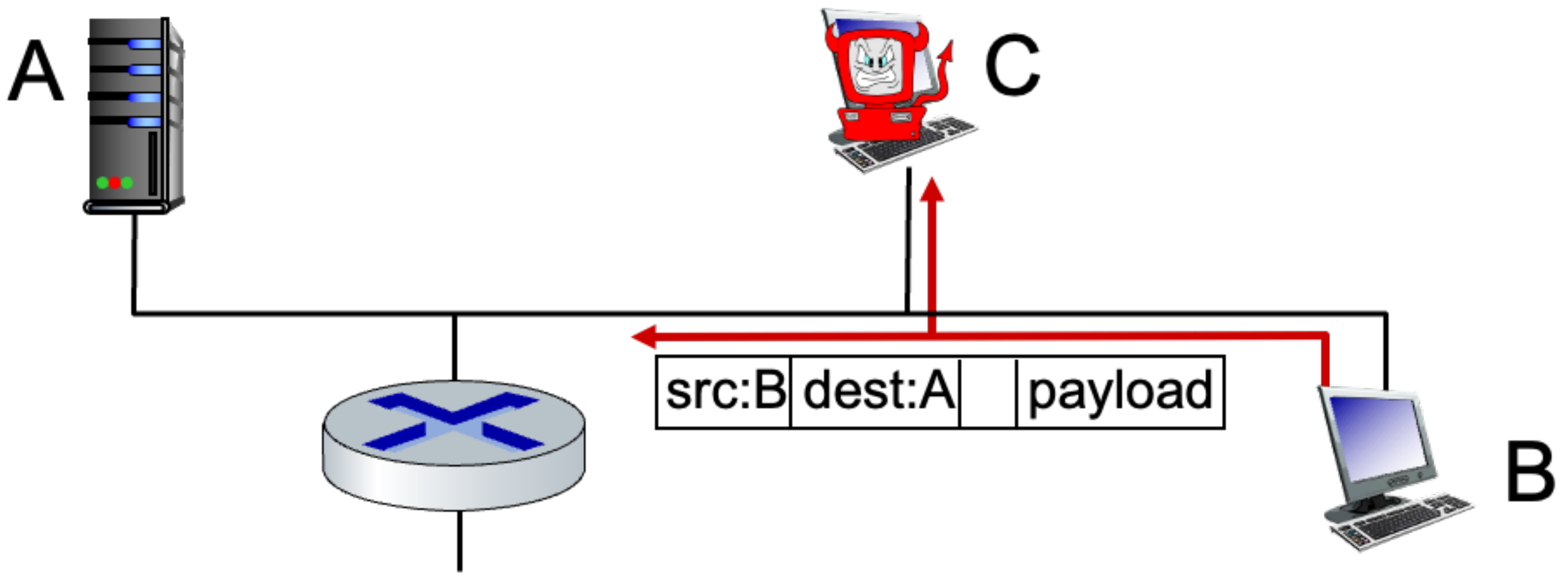
Security

Network security

- Internet was not originally designed with (much) security in mind
  - Original vision: “a group of mutually trusting users attached to a transparent network”
  - Internet protocol designers are playing “catch-up”
  - Security considerations are integrated across all layers!
- We now need to think about
  - How bad guys can attack computer networks
  - How we can defend computer networks against attacks
  - How to design architectures that are immune to attacks

Bad guys : packet interception

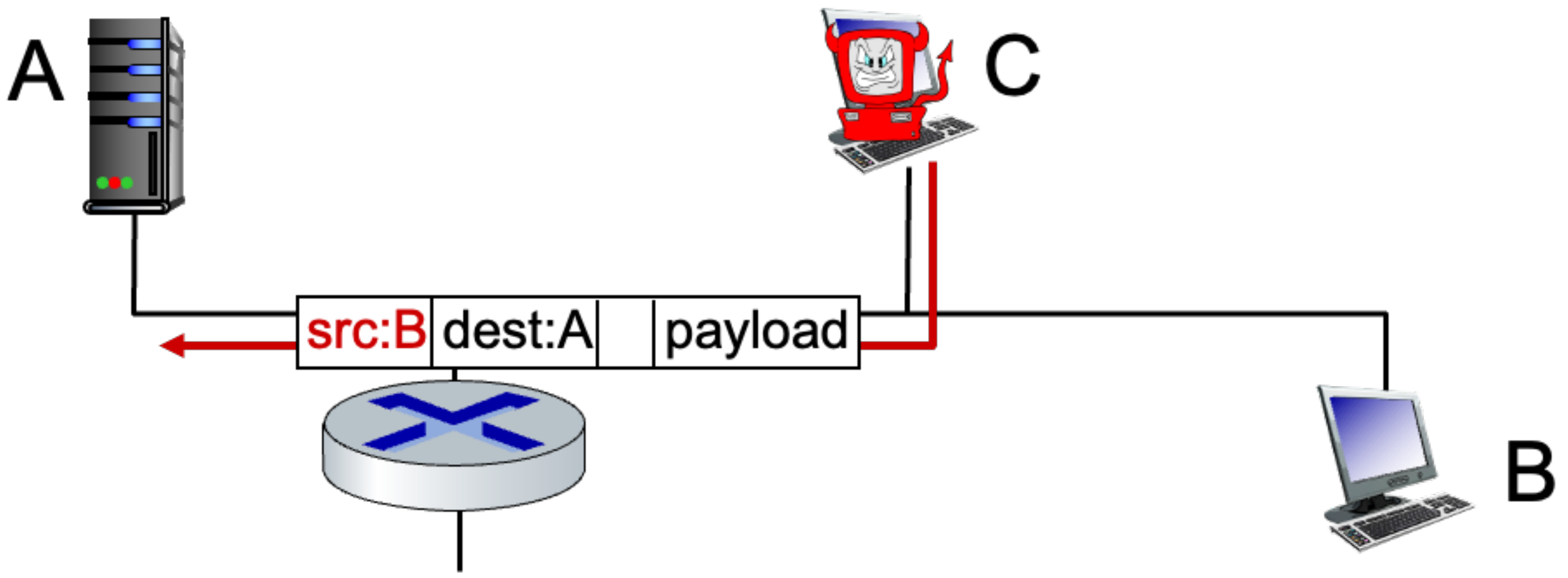
- **Packet "sniffing"**
  - Typically under broadcast media (shared ethernet, wireless)
  - Promiscuous network interface reads/records all packets (e.g., including passwords) passing by



Wireshark software used for our end-of-chapter labs is a (free) packet-sniffer

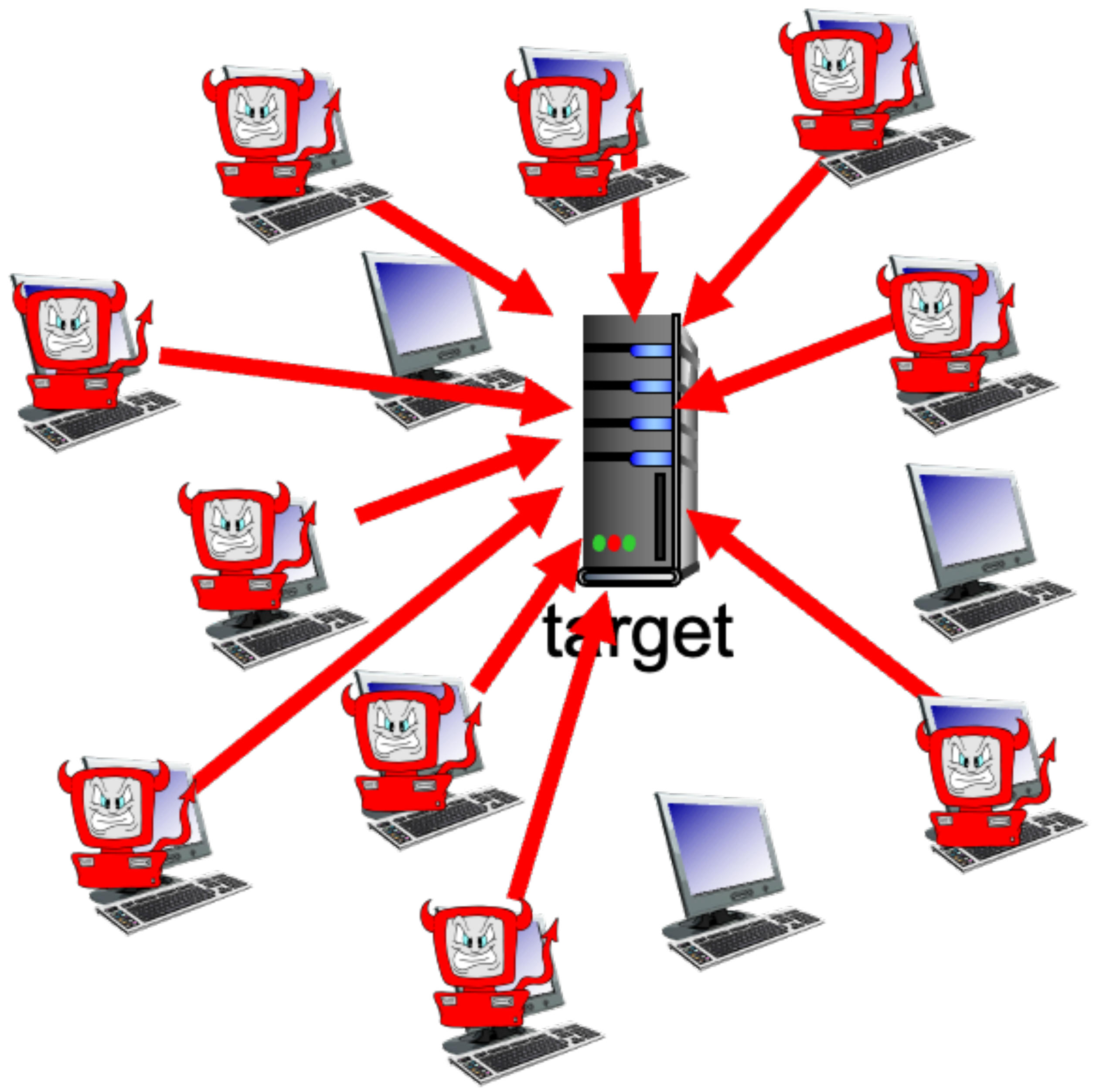
Bad guys : fake identity

- **IP spoofing**
  - Injection of packets with false source address



Bad guys : denial of service

- **Denial of Service (DoS)**
  - Attackers make resources (server, bandwidth) unavailable to legitimate traffic by overwhelming resource with bogus traffic
  - Attack procedure
    - Select target
    - Break into hosts
      around the network (see botnet)
    - Send packets to target from compromised host



Lines of defense

- **Authentication**: proving you are who you say you are
  - Cellular networks provide hardware identity via SIM card
  - No such hardware assists in the traditional Internet
- **Confidentiality**: via encryption
- **Integrity checks**: digital signature prevents/detects tempering
- **Access restrictions**: password-protected VPNs
- **Firewall**: specialized “middleboxes” in access and core networks
  - Off-by-default: filter incoming packets to restrict senders, receivers, applications
  - Detecting/reacting to DoS attacks

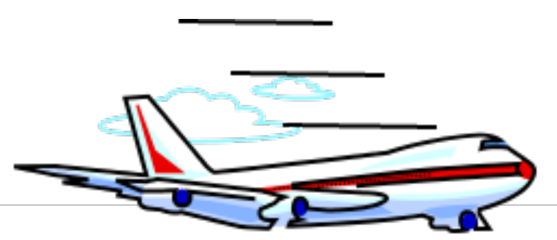
… lots more on security

Protocol layers, Service models

Protocol “Layers” and reference models

- Networks are complex, with many “pieces”
  - Hosts
  - Routers
  - Links of various media
  - Applications
  - Protocols
  - Hardware / software

Organization of air travel (analogy)



end-to-end transfer of person plus baggage

ticket (purchase)	ticket (complain)
baggage (check)	baggage (claim)
gates (load)	gates (unload)
runway takeoff	runway landing
airplane routing	airplane routing

airplane routing

How would you define/discuss the system of airline travel?
(Considering that it has series of steps, involving many services)

Organization of air travel (analogy)

Layers: each layer implements a service

→ via its own internal-layer actions

→ relying on services provided by layer below

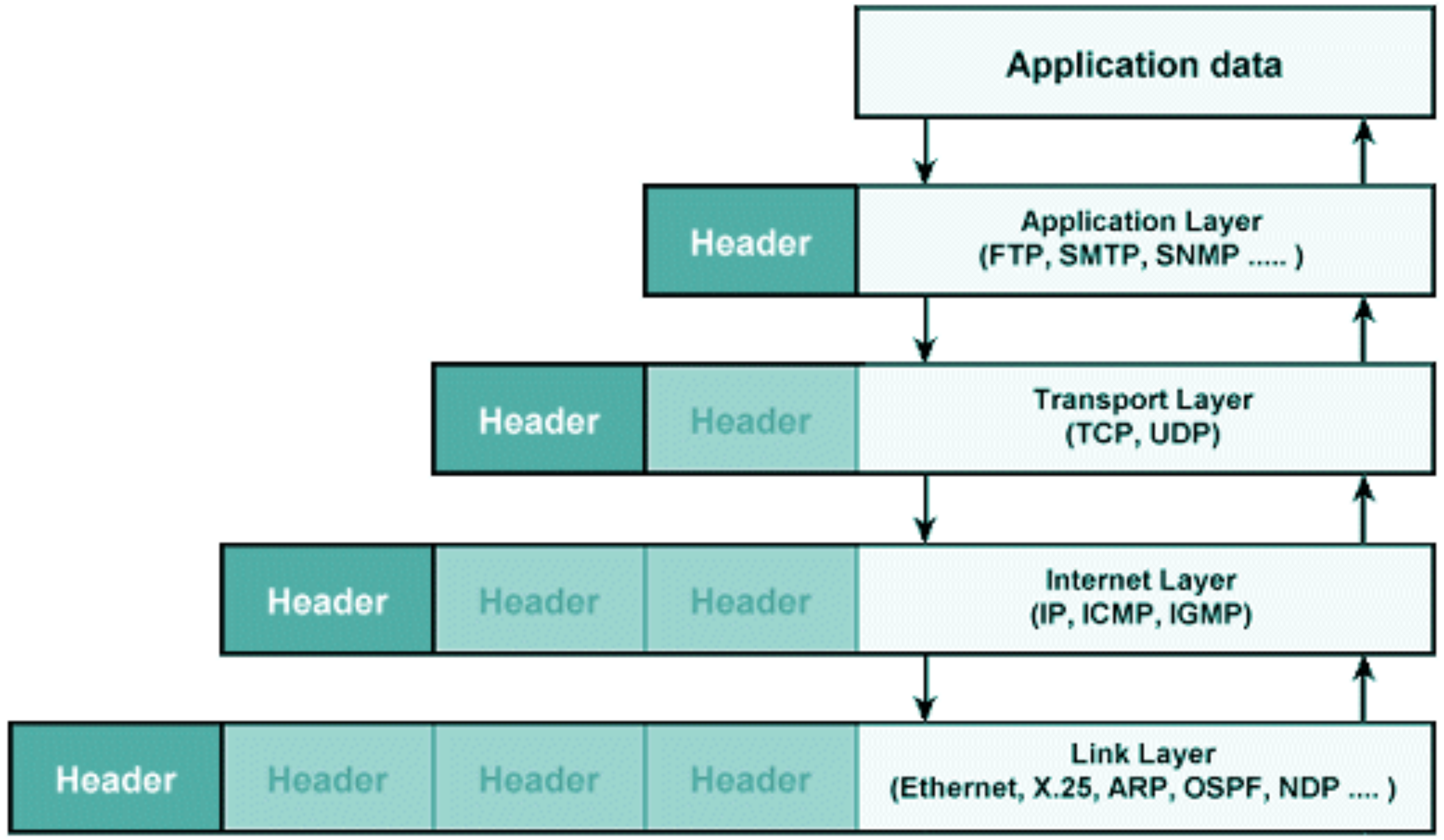
Why Layering?

- Approach to designing/discussing complex systems
  - Modularization eases maintenance, updating of a system
    - Change in layer's service implementation: transparent to rest of a system

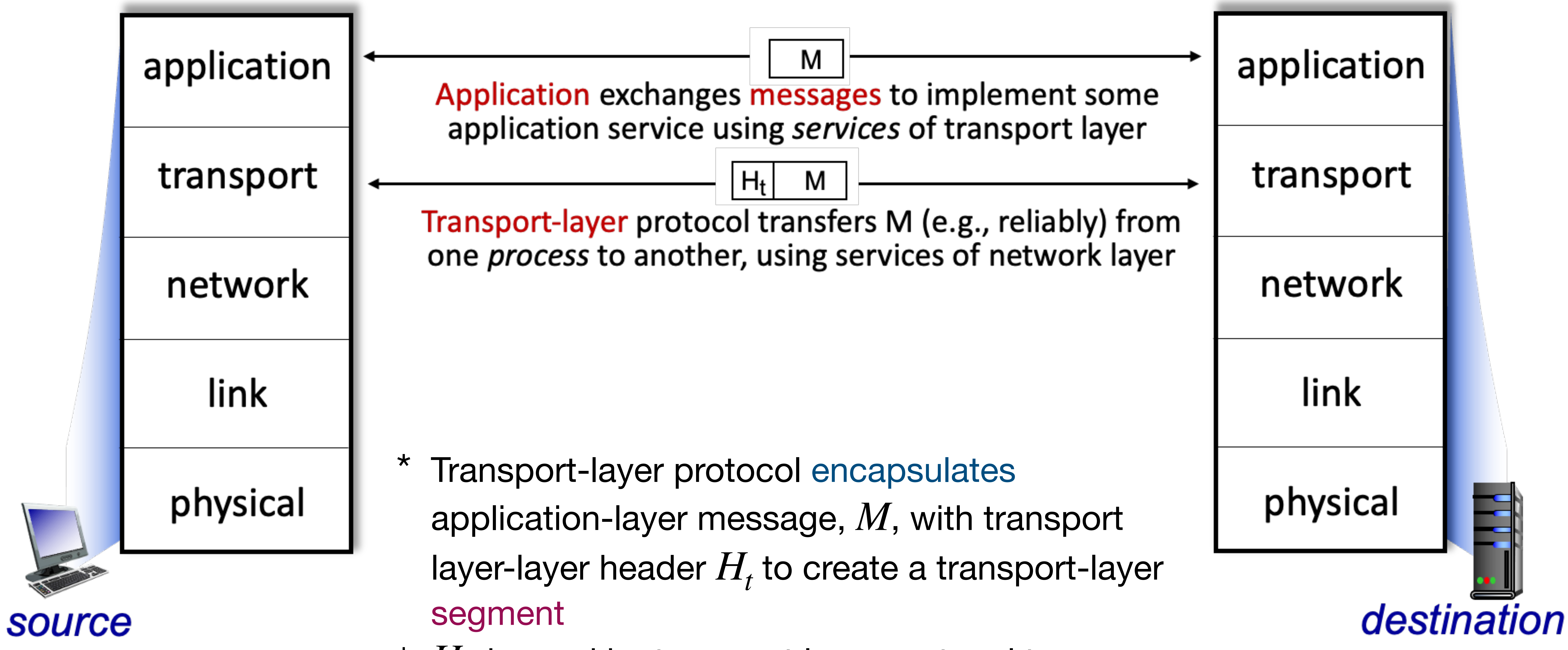
      e.g., change in gate procedure does not affect rest of the entire airline system
  - Explicit structure allows identification, relationship of system's pieces
    - Layered reference model for discussion

Layered Internet protocol stack

- Application: supporting network applications
  - HTTP, IMAP, SMTP, DNS
- Transport
  - TCP, UDP
- Network
  - IP, routing protocols
- Link
  - Ethernet, IEEE 802.11 (Wifi), PPP
- Physical: bits “on the wire”

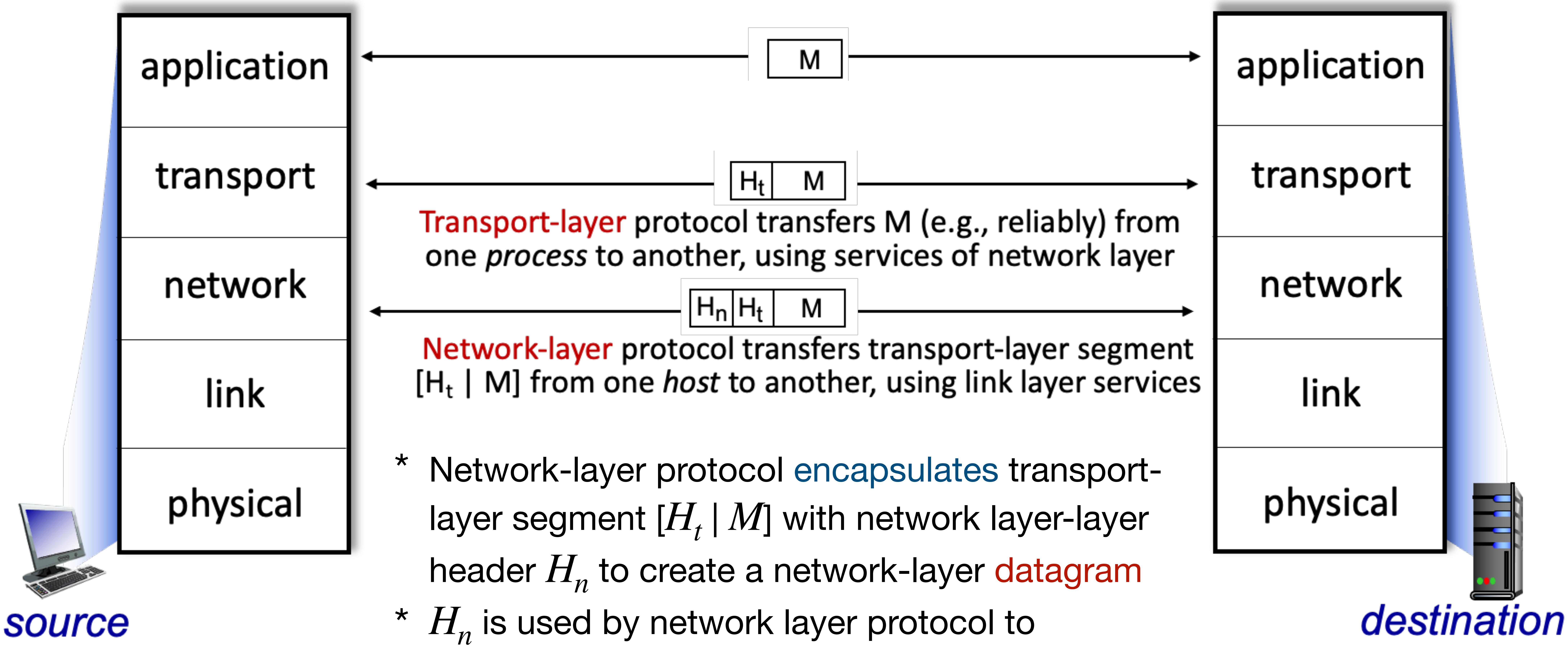


Services, Layering, and Encapsulation



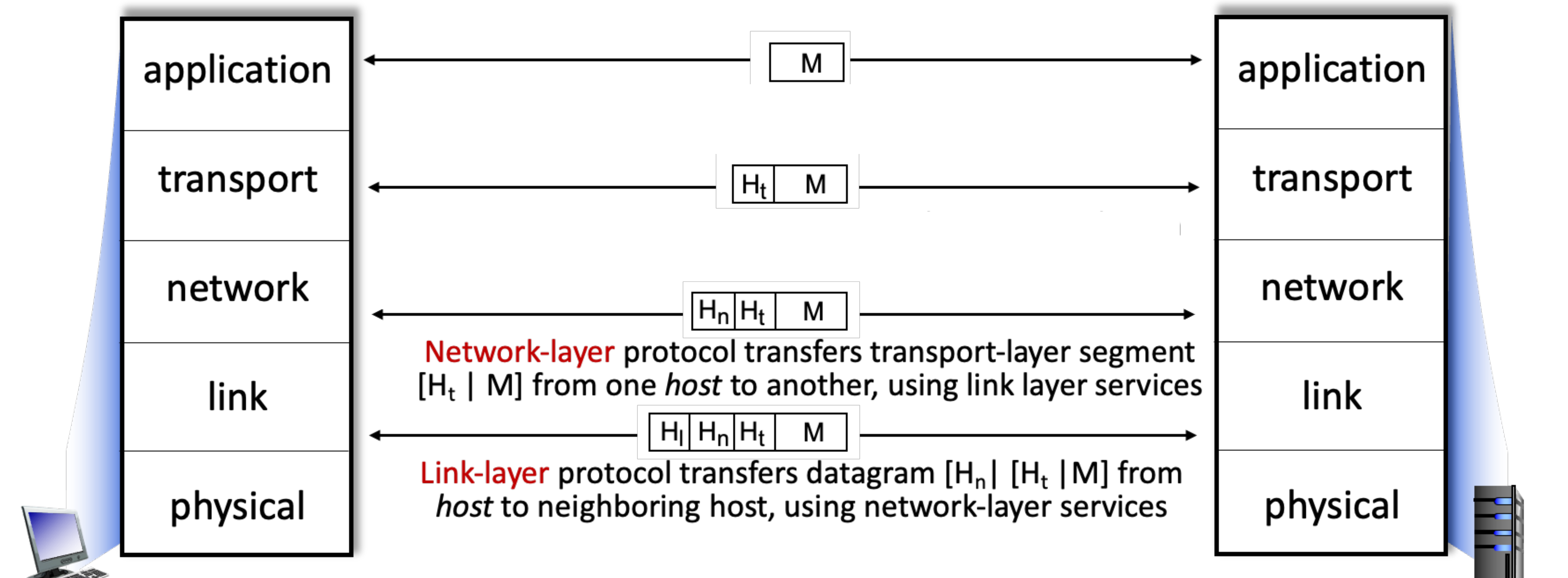
* Transport-layer protocol encapsulates application-layer message, M, with transport layer-layer header H_t to create a transport-layer segment
* H_t is used by transport layer protocol to implement its service

Services, Layering, and Encapsulation



- Network-layer protocol encapsulates transport-layer segment $[H_t \mid M]$ with network layer-layer header H_n to create a network-layer datagram
- H_n is used by network layer protocol to implement its services

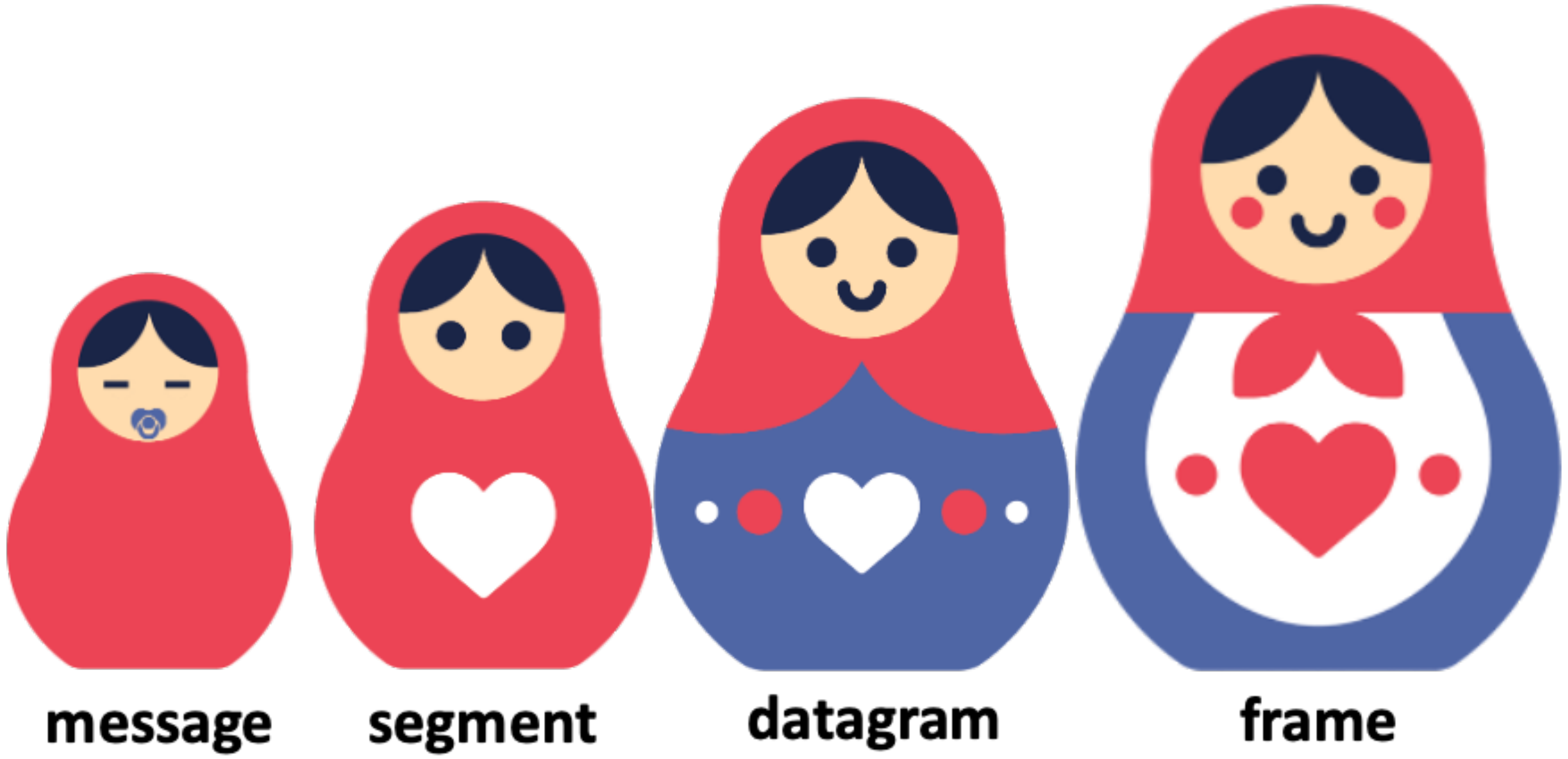
Services, Layering, and Encapsulation



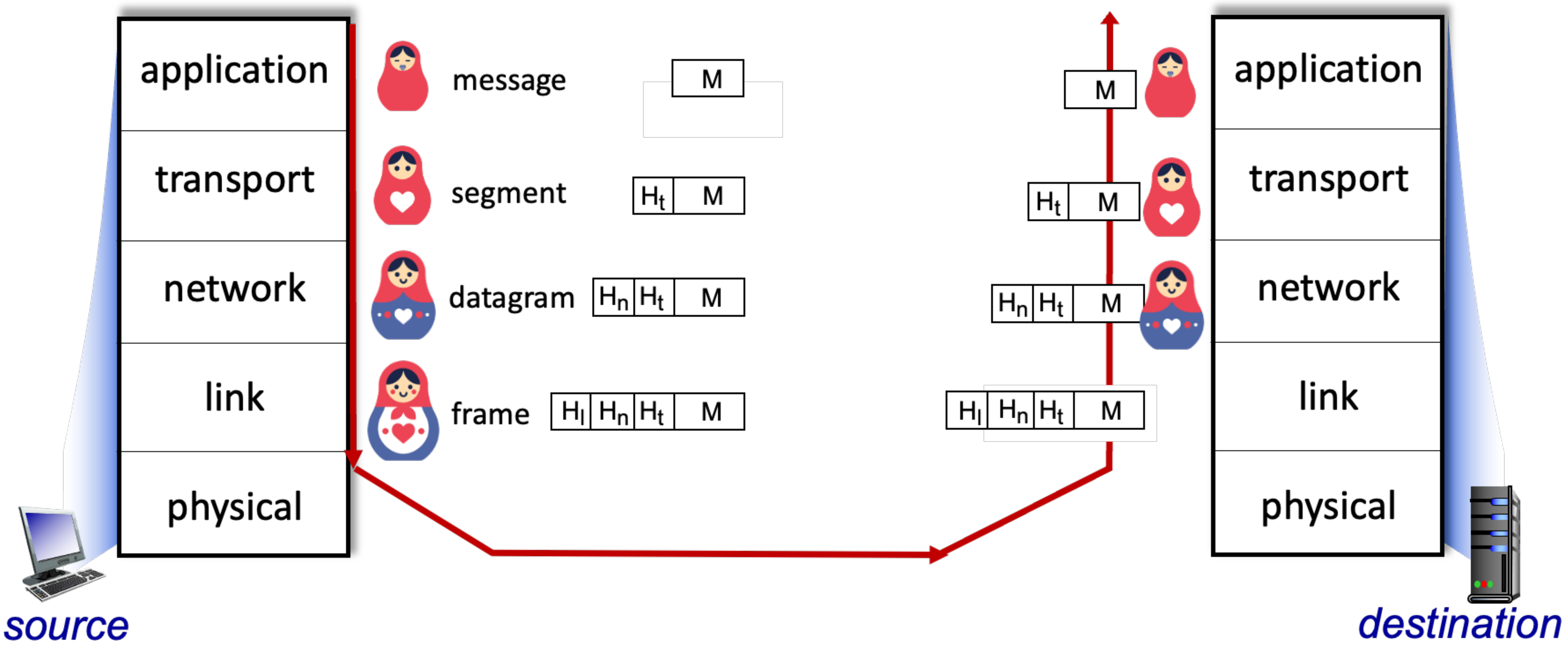
* Link-layer protocol encapsulates network-layer datagram $[H_n \mid H_t \mid M]$ with link layer-layer header H_l to create a link-layer frame

Encapsulation

Matryoshka dolls (stacking dolls)

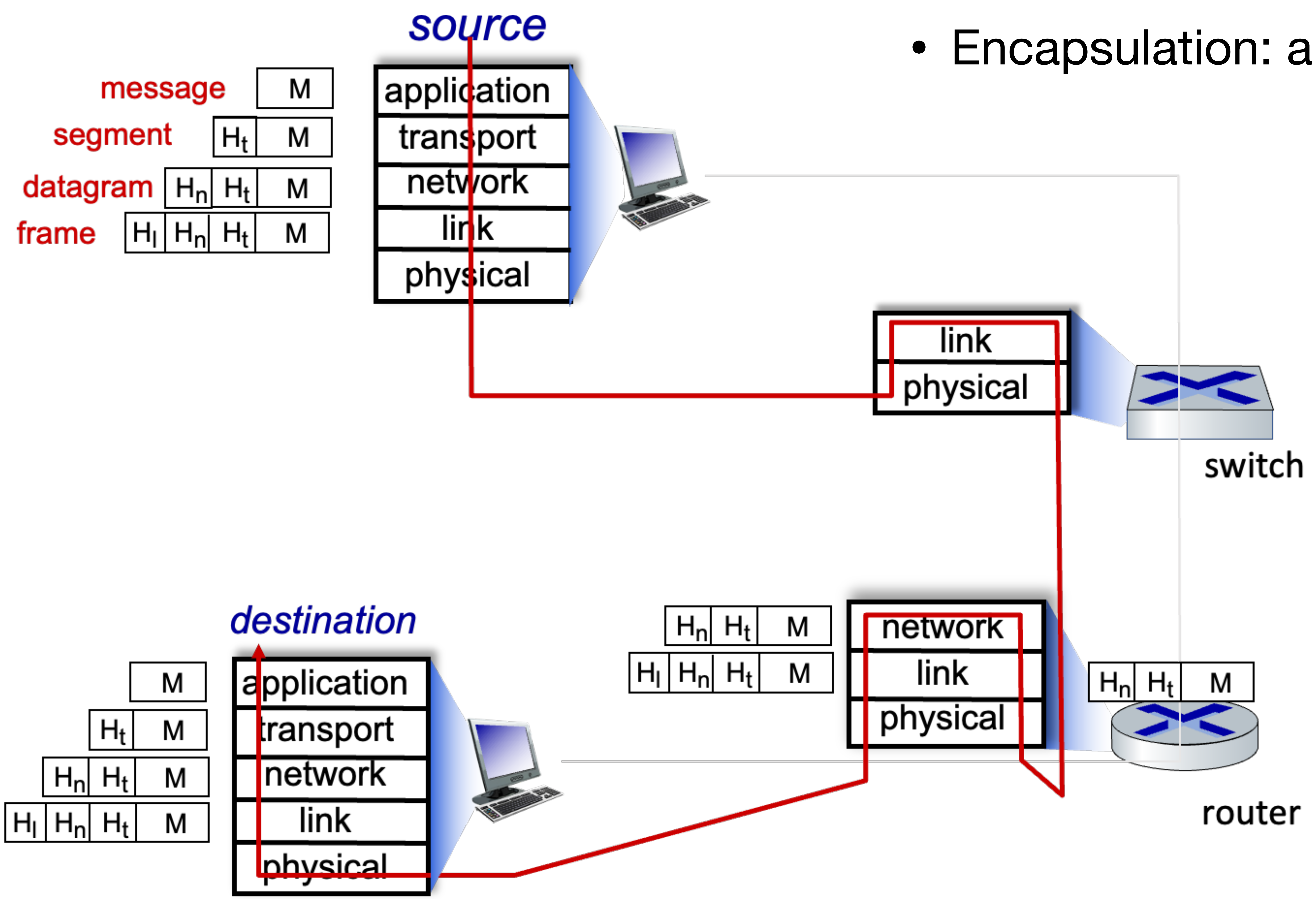


Services, Layering, and Encapsulation



Services, Layering, and Encapsulation

- Encapsulation: an end-to-end view



Questions?

Wireshark

- Used to capture packets on the wire